7 Digital Shift Keying Techniques

Objective

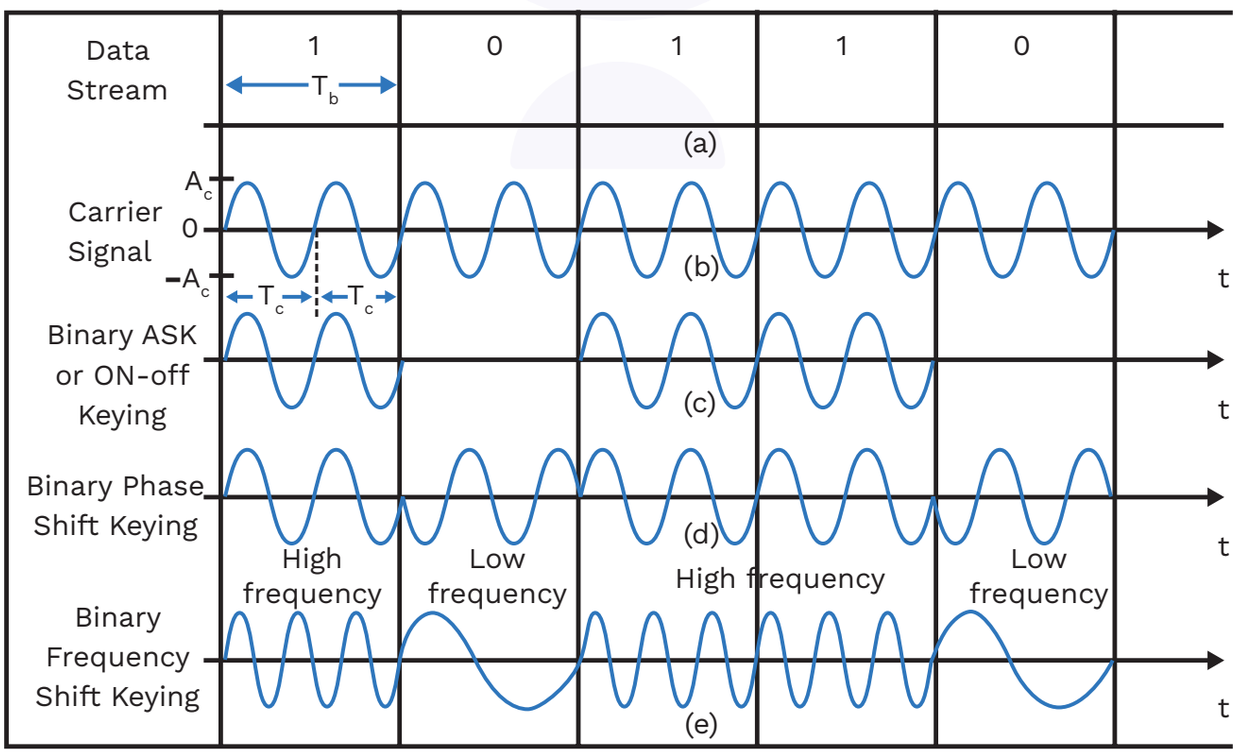
After completing this chapter user will be able to understand:

- What is bandpass modulation?
- Different types of modulation techniques.
- Constellation diagrams of modulation techniques.
- M-Ary bandpass modulation.

Introduction

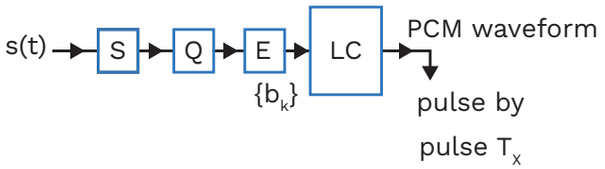
Digital modulation techniques of band-pass modulations are used for transmission through free space. The incoming message signal is modulated onto a carrier signal, which is usually sinusoidal; the modulation is done with fixed frequency limits imposed by the channel considered.

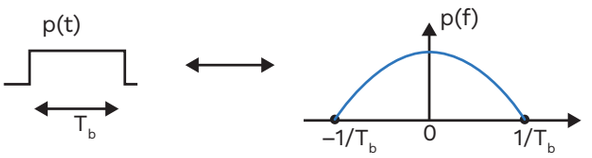
- The channel can be a microwave radio link, satellite channel, etc.
- For a digital message signal, the modulation process involves switching or keying the amplitude, frequency, or phase of the carrier.
- Thus, the three basic signalling schemes are Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Phase Shift Keying (PSK).
- These are similar to their analogue counterparts, i.e., Amplitude modulation, Frequency modulation and Phase modulation, respectively.
- The waveforms of these switching/keying techniques by defining a particular digital data (message signal) as an example to show all three techniques, BASK, BFSK and BPSK are listed below.



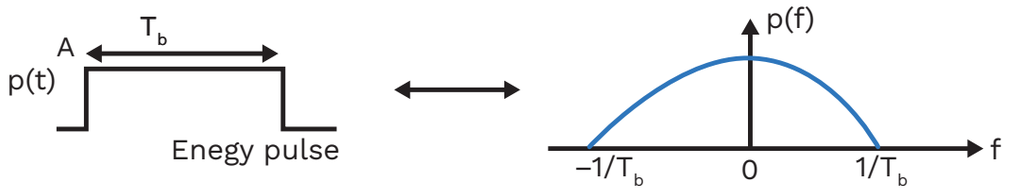
Bandpass Digital Communication

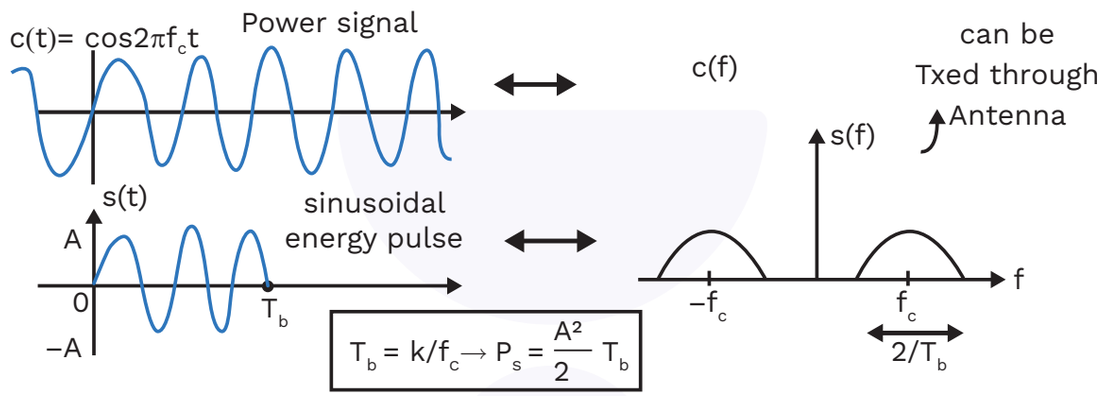
Baseband digital communication:





1. p(t) can be Txed through wired channel.
2. p(t) can't be Txed through wireless channel.





$$T_b = \frac{k}{f_c} \rightarrow P_s = \frac{A^2}{2} T_b$$

Important Points

1. In baseband digital communication, low pass energy pulse w.r.t. '1' and '0' is Txed.

$$s(t) = \begin{cases} s_1(t) = A & 0 \le t \le T_b : \ 1T_y \\ s_2(t) = -A & 0 \le t \le T_b : \ 0T_x \end{cases}$$

2. In bandpass digital communication, sinusoidal energy pulses are transmitted w.r.t '1' and "0"

a) $$s(t) = \begin{cases} s_1(t) A & 0 \le t \le T_b : \ 1T_y \\ s_2(t) = -A & 0 \le t \le T_b : \ 0T_x \end{cases}$$

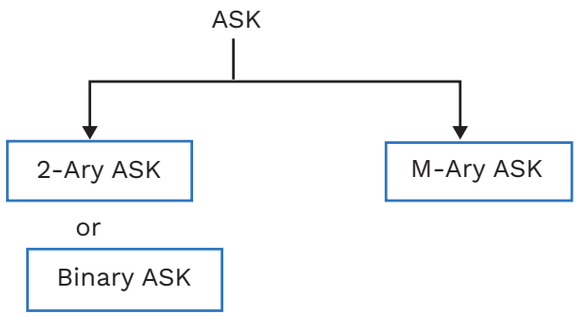
s (t) → Amplitude shift keying ↔On-off keying waveform (ASK).

b) $$s(t) = \begin{cases} s_1(t) A\cos(\omega_c t + \theta_1) & 0 \le t \le T_b : \ 1T_x \\ s_2(t) = A\cos(\omega_c t + \theta_2) & 0 \le t \le T_b; \ 0T_x \end{cases}$$

Phase shift keying → PSK

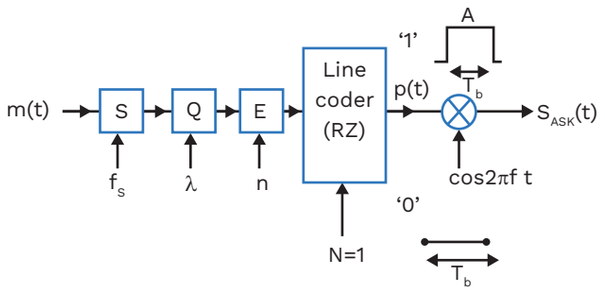
c) $$s(t) = \begin{cases} s_1(t) A\cos(2\pi f_1 t) & 0 \le t \le T_b : \ 1T_x \\ s_2(t) = A\cos(2\pi f_2 t) & 0 \le t \le T_b; \ 0T_x \end{cases}$$

Frequency shift keying→ FSK



Binary Ask

Tx:

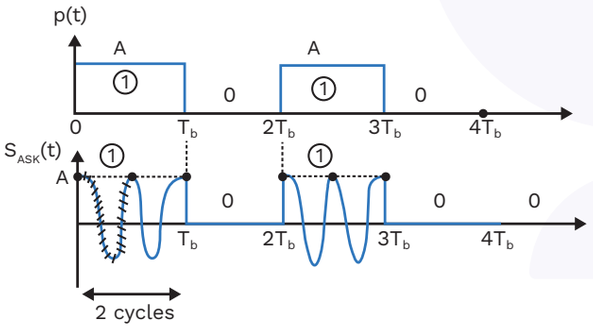


$$p(t)=\begin{cases}A & 0\le t\le T_b: \ 1T_x\\ 0 & 0\le t\le T_b: \ 0T_x\end{cases}$$

$$s(t)=\begin{cases}s_1(t)=A\cos 2\pi f_c t & : \ 0\le t\le T_b \quad 1T_x\\ s_2(t)=0\cos 2\pi f_c t & : \ 0\le t\le T_b \quad 0T_x\end{cases}$$

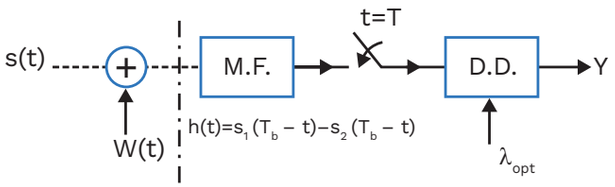
$$\{b_k\}=\{1,\ 0,\ 1,\ 0\}$$

Let $T_b=\dfrac{k}{f_c}$





Matched Filter Rx



$$s(t)=\begin{cases}s_1(t)=A\cos 2\pi_c t & 0\le t\le T_b\\ s_2(t)=0 & 0\le t\le T_b\end{cases}$$

1. $h(t)=s_1(T_b-t)-s_2(T_b-t)\to 0$

$=A\cos\left[2\pi f_c T_b-2\pi f_c t\right] \qquad f_c T_b\in k$

$h(t)=A\cos\left[2\pi f_c t\right] \qquad 0\le t\le T_b$

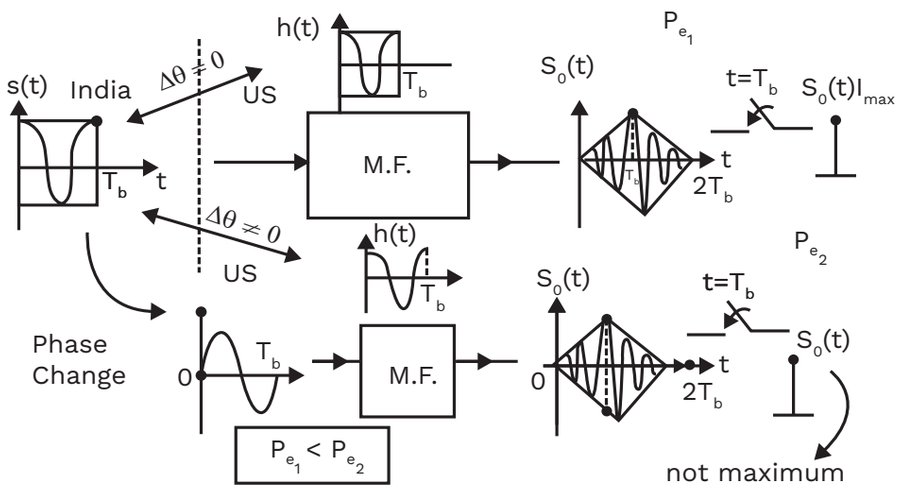
2. $P_e=Q\left[\sqrt{\dfrac{E_d}{2N_0}}\right]$

$$E_d=\int_0^{T_b}\left[s_1(t)-s_2(t)\right]^2 dt=\int_0^{T_b}s_1^2(t)\,dt$$

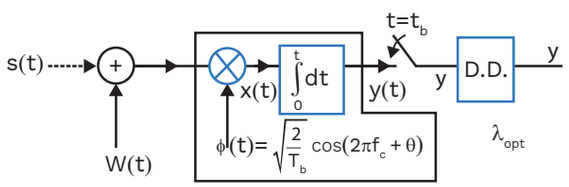
$If\ f_c=k/T_b \qquad E_d=\dfrac{A^2}{2}xT_b$

$$(P_e)_{ASK}=Q\left[\sqrt{\frac{A^2T_b}{4N_0}}\right]$$

Problem:



Correlator receiver: $T_b = k / f_c$



$$s(t)=\begin{cases} A\cos 2\pi f_c t & : \ 0\le t\le T_b \\ 0 & else \end{cases}$$

$$M=2\rightarrow N=1$$

$$\phi(t)=\begin{cases}\sqrt{\frac{2}{T_b}}\cos 2\pi f_c t & 0\le t\le T_b \\ 0 & else\end{cases}$$

$$\text{So, } \phi(t)=\begin{cases}\sqrt{\frac{2}{T_b}}\cos(2\pi f_c t+\theta) & 0\le t\le T_b \\ 0 & else\end{cases}$$

θ : Phase diff between Txed energy pulse carrier and locally generated energy pulse carrier at Rx.

$$s(t)+W(t)=\begin{cases}A\cos 2\pi f_c t+W(t) & 0\le t\le T_b \quad 1T_x \\ 0+W(t) & 0\le t\le T_b \quad 0T_x\end{cases}$$

$$x(t)=\begin{cases}A\cos 2\pi f_c t\phi(t)+W(t)\phi(t) & : \ 1T_x \\ W(t)\phi(t) & : \ 0T_x\end{cases}$$

$$Y=\begin{cases}\int_0^{T_b} A\cos 2\pi f_c t\phi(t)dt+\overset{N}{\int_0^{T_b}} W(t)\phi(t)dt & 1T_x \\ \int_0^{T_b} W(t)\underset{N}{\phi(t)}dt & 0T_x\end{cases}$$

$$N: GRV=\begin{cases}E[N]=0 \\ \sigma_N^2=\frac{N_0}{2}\end{cases}$$

$$=\int_0^{T_b}\frac{\sqrt{2}A}{\sqrt{T_b}}\cos 2\pi f_c t\cos(2\pi f_c t+\theta)dt$$

$$=\int_0^{T_b}\frac{\sqrt{2}A}{2\sqrt{T_b}}\overset{0}{\cos}(4\pi f_c t+\theta)dt+\int_0^{T_b}\frac{\sqrt{2}A}{2\sqrt{T_b}}\cos\theta dt$$

$$=\frac{A}{\sqrt{2}}\sqrt{T_b}\cos\theta$$

$$Y=\begin{cases}\frac{A\sqrt{T_b}}{\sqrt{2}}\cos\theta+N & 1T_x \\ N & 0T_x\end{cases}$$

$$E[Y]=\begin{cases}\frac{A\sqrt{T_b}}{\sqrt{2}}\overset{\mu_1}{\cos\theta} & 1T_x \\ \underset{\mu 2}{0} & 0T_x\end{cases}$$

$$\sigma_Y^2=\begin{cases}\sigma_N^2=N_0/2: & 1T_x \\ \sigma_N^2=N_0/2: & 0T_x\end{cases}$$

$$P_e\big|_{\min}\big|_{\min}=Q\left[\frac{\mu_1-\mu_2}{2\sigma_y}\right]\rightarrow \mu_1,\mu_2,\sigma_4 : correlator.$$

$$P_e=Q\left[\sqrt{\frac{A^2T_b}{4N_0}}\cos^2\theta\right]\sigma_y=\sqrt{\frac{N_0}{2}}$$

$$0\le\cos^2\theta\le 1 \qquad P_e(\theta\neq 0)>P_e(\theta=0)$$

$$(P_e)_{ASK}=Q\left[\sqrt{\frac{A^2T_b}{4N_0}}\cos^2\theta\right]$$

$$\lambda_{0pt}=\frac{\mu_1+\mu_2}{2}+\frac{\sigma_y^2}{\mu_1-\mu_2}\ell n\left\{\frac{\overset{0}{P(0T_x)}}{P(1T_x)}\right\}$$

$$\lambda_{0pt}=\frac{A}{2}\sqrt{\frac{T_b}{2}}\cos\theta$$

Geometrical Representation of BASK

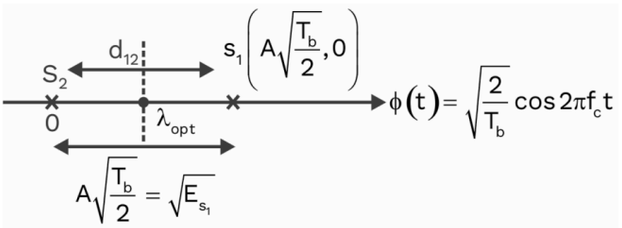
Case 1: $\theta=0$

$$s(t)=\begin{cases}A\cos 2\pi f_c t: & 0\le t\le T_b: \quad 1T_x \\ 0 & 0\le t\le T_b: \quad 0T_x\end{cases}$$

$$\phi(t)=\begin{cases}\sqrt{\frac{2}{T_b}}\cos 2\pi f_c t & 0\le t\le T_b \\ 0 & else\end{cases}$$

$$s(t)=\begin{cases}A\sqrt{\frac{T_b}{2}}\phi(t)\rightarrow s_1: & 0\le t\le T_b: \quad 1T_x \\ 0\phi(t)\rightarrow s_2 & 0\le t\le T_b: \quad 0T_x\end{cases}$$

Constellation:



$$P_e = Q\left[\sqrt{\frac{d_{12}^2}{2N_0}}\right] = Q\left[\sqrt{\frac{A^2T_b}{4N_0}}\right] \quad \lambda_{opt} = \frac{A\sqrt{T_b/2}}{2}$$

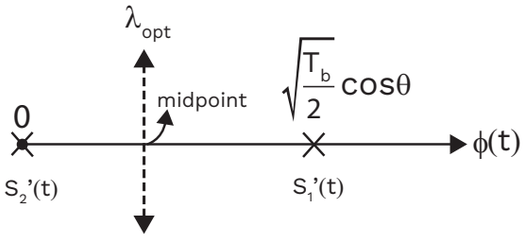
Case 2:

$$\theta \neq 0 \qquad s(t) = \begin{cases} s_1(t) = A\cos 2\pi f_c t & 1T_x \\ s_2(t) = 0 & 0T_x \end{cases}$$

$$\phi(t) = \begin{cases} \sqrt{\frac{2}{T_b}}\cos(2\pi f_c t + \theta): & 0 \le t \le T_b \\ 0: & else \end{cases}$$

$$P_e = Q\left[\sqrt{\frac{A^2T_b}{EN_0}\cos^2\theta}\right]$$

Modified Constellation Diagram



$$d_{12} = A\sqrt{\frac{T_b}{2}}\cos\theta \rightarrow P_e$$

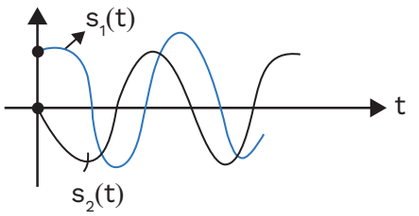
$$\lambda_{opt} = \frac{A\sqrt{\frac{T_b}{2}}\cos\theta}{2}$$

Phase Orthogonal (Vs) Space Orthogonal

1. **Phase orthogonal:**

$$s_1(t) = A\cos\omega_c t$$

$$s_2(t) = A\cos(\omega_c t + 90°)$$



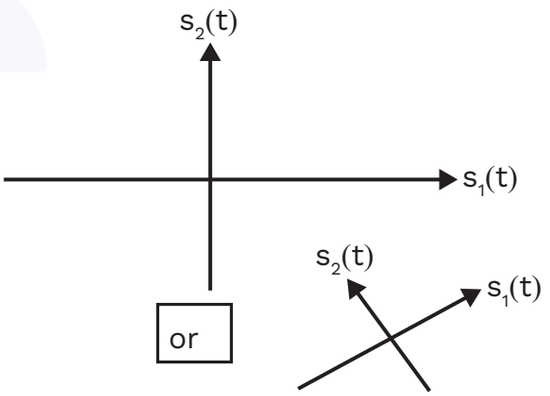
2. **Space orthogonal:**

$$s_1(t) = \sqrt{\frac{2}{T_b}}\cos\omega_c t \qquad 0 \le t \le T_b$$

$$s_2(t) = \sqrt{\frac{2}{T_b}}\sin\omega_c t \qquad 0 \le t \le T_b$$

$$T_b = k / f_c$$

$$\begin{cases} \|s_1(t)\| = \int_0^{T_b} s_1(t).s_1(t)\,dt = 1 \\ \|s_2(t)\| = \int_0^{T_b} s_2(t)s_2(t)\,dt = 1 \\ < s_1(t).s_2(t) > = \int_0^{T_b} s_1(t)s_2(t)\,dt = 0 \end{cases}$$



Example:

$$s(t) = \begin{cases} A\cos(\omega_c t + \alpha) & 0 \le t \le T_b \quad 1T_x \\ 0 & 0 \le t \le T_b \quad 0T_x \end{cases}$$

"Draw constellation diagram of above Band-pass digitally modulated signal."

Solution:

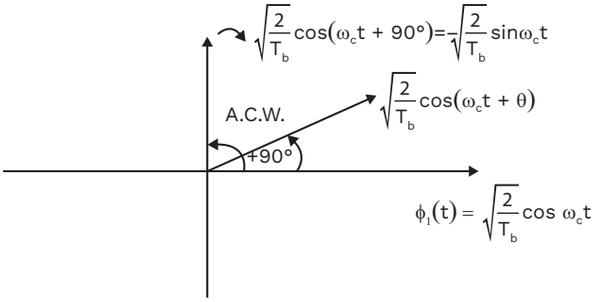
$$s(t) = A\cos(\omega_c t + \alpha): \qquad 0 \le t \le T_b$$

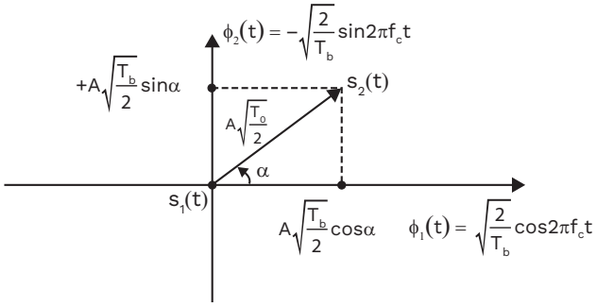
$$s(t) = A\cos(\omega_c t)\cos\alpha - A\sin\omega_c t \sin\alpha : 0 \le t \le T_b$$

$$s(t) = A\cos\alpha\sqrt{\frac{T_b}{2}}\left\{\sqrt{\frac{2}{T_b}}\cos\omega_c t\right\} + \left(A\sin\alpha\right)\sqrt{\frac{T_b}{2}}\left\{-\sqrt{\frac{2}{T_b}}\sin\omega t\right\} \quad 0 \le t \le T_b$$

$\phi_1(t)$ $\phi_2(t)$

$$s(t) = \left\{A\sqrt{\frac{T_b}{2}}\cos\alpha\right\}\phi_1(t) + \left\{A\sqrt{\frac{T_b}{2}}\sin\alpha\right\}\phi_2(t) \qquad 0 \le t \le T_b$$

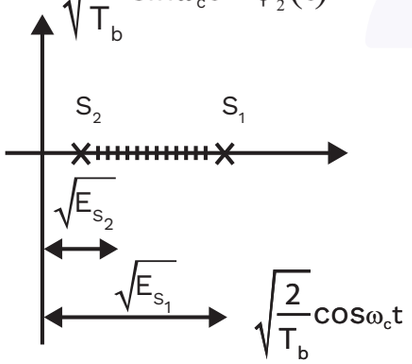




$\phi_1(t), \phi_2(t) \rightarrow$ *set of orthonomal basis function.*

$$s(t) = \begin{cases} 2A\cos\omega_c t : 0 \le t \le T_b \\ A\cos\omega_c t : 0 \le t \le T_b \end{cases}$$

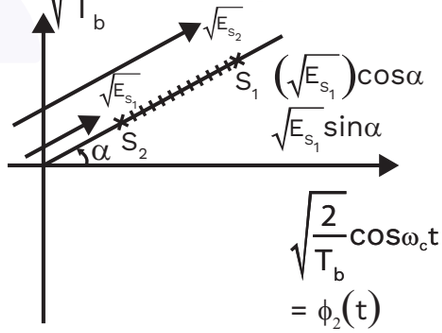
$$-\sqrt{\frac{2}{T_b}}\sin\omega_c t = \phi_2(t)$$



$$\sqrt{\frac{2}{T_b}}\cos\omega_c t = \phi_2(t)$$

$$s(t) = \begin{cases} 2A\cos(\omega_c t + \alpha): 0 \le t \le T_b \\ A\cos(\omega_c t + \alpha): 0 \le t \le T_b \end{cases}$$

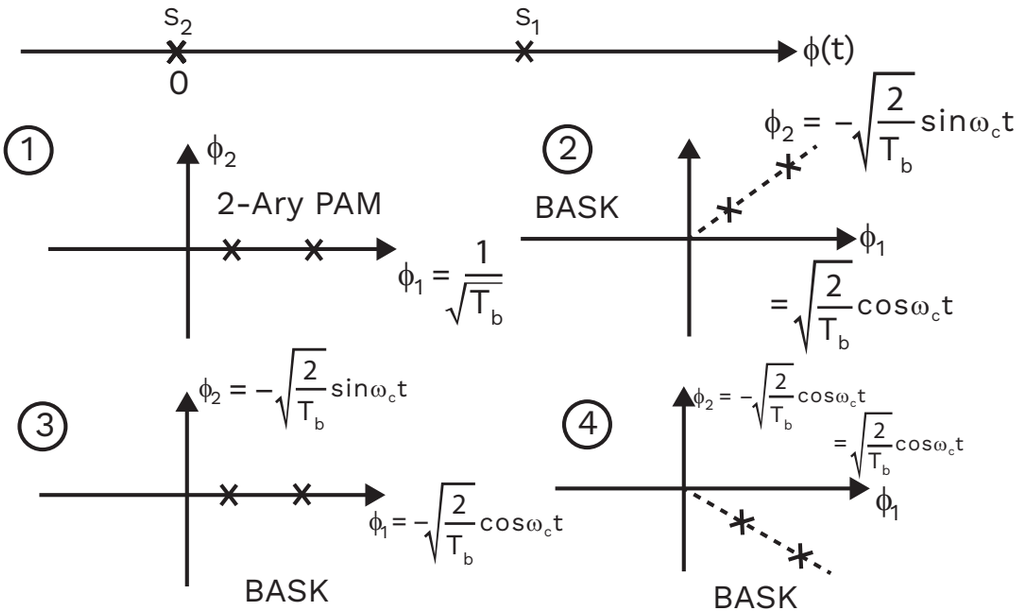
$$-\sqrt{\frac{2}{T_b}}\sin\omega_c t = \phi_2(t)$$



$$\sqrt{\frac{2}{T_b}}\cos\omega_c t = \phi_2(t)$$

$$s(t) = \begin{cases} A\cos(\omega_c t + \alpha) & 0 \le t \le T_b \\ 0 & else \end{cases}$$

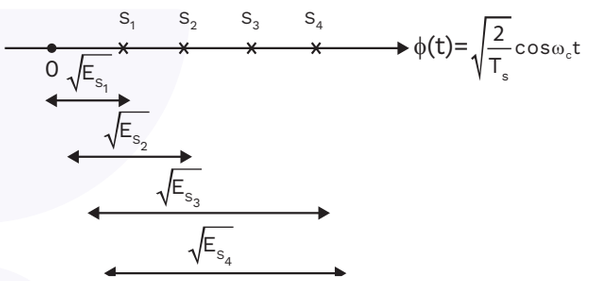
$$\phi(t) = \begin{cases} \sqrt{\frac{2}{T_b}}\cos(\omega_c t + \alpha): & 0 \le t \le T_b \\ 0 & else \end{cases}$$



M-Ary Ask

4-Ary Ask: M=4 N = 2 → duration $2T_b = T_s$

$$s(t)\begin{cases} s_1(t) = A\cos\omega_c t : 0 \le t \le T_s \quad "00" \\ s_2(t) = 2A\cos\omega_c t : 0 \le t \le T_s \quad "01" \\ s_3(t) = 3A\cos\omega_c t : 0 \le t \le T_s \quad "10" \\ s_4(t) = 4A\cos\omega_c t : 0 \le t \le T_s \quad "11" \end{cases}$$

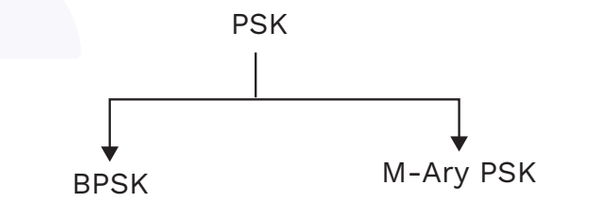


$T_s = k / f_c$

$E_{s_1} \ne E_{s_2} \ne E_{s_3} \ne E_{s_4} \rightarrow "M-Ary\ Ask\ is$ *not* constant energy scheme."

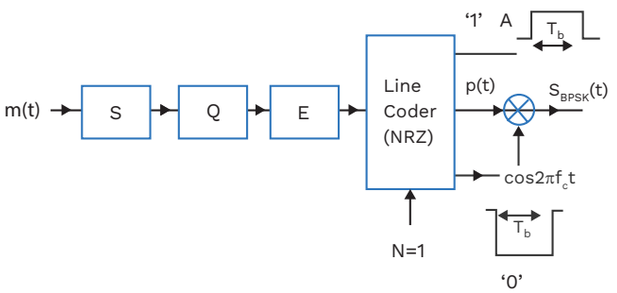
$$\phi(t) = \begin{cases} \sqrt{\frac{2}{T_s}}\cos\omega_c t : & 0 \le t \le T_s \\ 0 : & elsewhere \end{cases}$$

$$s(t) = \begin{cases} A\sqrt{\frac{T_s}{2}}\phi(t) : 0 \le t \le T_s & 00 \\ 2A\sqrt{\frac{T_s}{2}}\phi(t) : 0 \le t \le T_s & 01 \\ 3A\sqrt{\frac{T_s}{2}}\phi(t) : 0 \le t \le T_s & 10 \\ 4A\sqrt{\frac{T_s}{2}}\phi(t) : 0 \le t \le T_s & 11 \end{cases}$$



Binary PSK

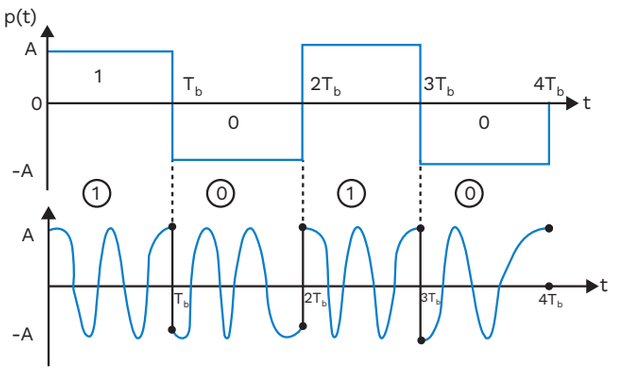
Tx



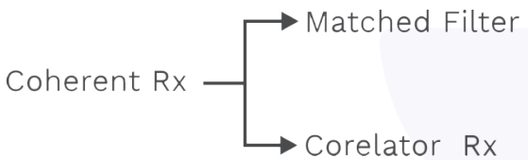
$$p(t) = \begin{cases} A & 0 \le t \le T_b : 1T_x \\ -A & 0 \le t \le T_b; 0T_x \end{cases}$$

$$s(t)=\begin{cases} A\cos 2\pi f_c t & 0\le t\le T: \quad 1T_x \\ A\cos\left(2\pi f_c t+\pi\right) & 0\le t\le T: \quad 0T_x \end{cases}$$

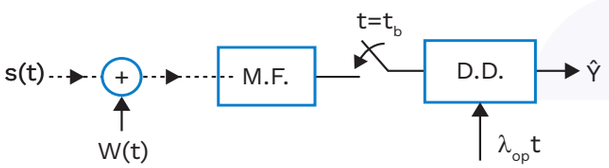
$$\{b_k\}=\{1,0,1,0\}$$



BPSK Rx



Matched Filter Based Rx



1. $h(t)=s_1(T_b-t)-s_2(T_b-t)$

$h(t)=2A\cos 2\pi f_c t: 0\le t\le T_b$

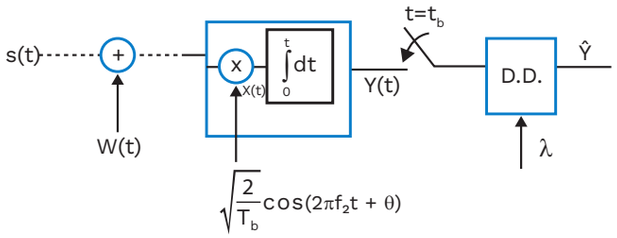
2. $P_e=Q\left[\sqrt{\dfrac{E_d}{2N_0}}\right]s_1(t)-s_2(t)$

$=2A\cos 2\pi f_c t \qquad 0\le t\le T_b$

$$E_d=\int_0^{T_b}\left(2A\cos 2\pi f_c t\right)^2 dt=\frac{4A^2}{2}\times T_b=2A^2T_b.$$

$$\left(P_e\right)_{BPSK}=Q\left[\sqrt{\frac{A^2T_b}{N_0}}\right]$$

Correlator Based Rx



$$s(t)=\begin{cases} A\cos 2\pi f_c t & 0\le t\le T_b \quad 1T_x \\ -A\cos 2\pi f_c t & 0\le t\le T_b \quad 0T_y \end{cases}$$

$$\phi(t)=\begin{cases}\sqrt{\dfrac{2}{T_b}}\cos 2\pi f_c t \\ 0\le t\le T_b\end{cases}$$

$$x(t)=\begin{cases} A\cos 2\pi f_c t\phi(t)+W(t)\phi(t) & 1T_x \\ -A\cos 2\pi f_c t\phi(t)+W(t)\phi(t) & 0T_x \end{cases}$$

$$Y=\begin{cases} \int_0^{T_b} A\cos 2\pi f_c t\phi(t)dt+\overset{N}{\int_0^{T_b}} W(t)\phi(t)dt & 1T_y \\ -\int_0^{T_b} A\cos 2\pi f_c t\phi(t)dt+\int_0^{T_b} W(t)\phi(t)dt & 0T_x \end{cases}$$

$$Y=\begin{cases} A\sqrt{\dfrac{T_b}{2}}\cos\theta+N & 1T_x \\ -A\sqrt{\dfrac{T_b}{2}}\cos\theta+N & 0T_x \end{cases}$$

$$E(Y)=\begin{cases} A\sqrt{\dfrac{T_b}{2}}\overset{\mu_1}{\cos\theta}: & 1T_X \\ -A\sqrt{\dfrac{T_b}{2}}\underset{\mu_2}{\cos\theta}: & 0T_x \end{cases}$$

$$\sigma_Y^2=\begin{cases} \overset{\sigma_1^2}{N_0/2}: & 1T_x \\ N_0/2: & 0T_x \\ \underset{\sigma_2^2}{} \end{cases}$$

$$P_e\big|_{\min}=Q\left[\frac{\mu_1-\mu_2}{2\sigma_Y}\right]$$

$$P_e=Q\left[\sqrt{\frac{A^2T_b}{N_0}\cos^2\theta}\right]$$

$$\sigma_Y = \sqrt{\frac{N_0}{2}}$$

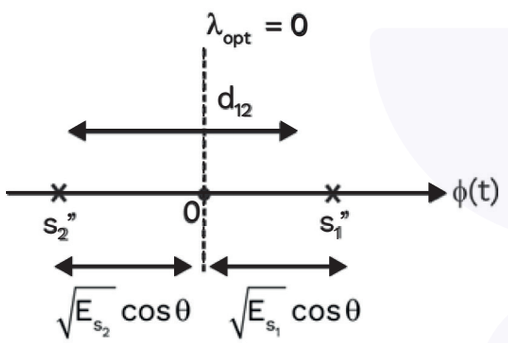
$$\lambda_{opt} = \frac{\mu_1 + \mu_2}{2} + 0 = 0$$

Geometrical Representation of BPSK

Case 1: $\theta = 0$

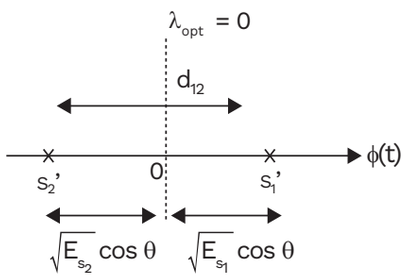
$$s(t) = \begin{cases} A\sqrt{\frac{T_b}{2}}\phi(t) & : \ 0 \le t \le T_b \ : \ 1T_x \\ -A\sqrt{\frac{T_b}{2}}\phi(t) & : \ 0 \le t \le T_b \ : \ 0T_x \end{cases}$$

$$P_e = Q\left[\sqrt{\frac{d_{12}^{\ 2}}{2N_0}}\right]$$



Case 2: Modified constellation diagram.

$$P_e = Q\left[\frac{d_{12}^{\ 2}}{2N_0}\right]$$



Binary PSK:

$$s(t) = \begin{cases} \overset{s_1(t)}{A\cos 2\pi f_c t} & 0 \le t \le T_b \quad 1T_x \\ \underset{s_2(t)}{-A\cos 2\pi f_c t} & 0 \le t \le T_b \quad 0T_x \end{cases}$$

$E_{s_1} = E_{s_2} \rightarrow$ BPSK is constant energy scheme.

M-Ary PSK

$\{b_K\} = \{.....0\,1\,1\,0\,1\,0.....\}$

→ N bits are grouped together so that $M = 2^N$ symbols or sinusoid energy pulse of duration $T_s = NT_b$ are generated: having the same A, same f and different phase.

M-Ary PSk

$$S_K(t) = A\cos\left(2\pi f_c t + \frac{2\pi}{M}K\right) \ 0 \le t \le T_s$$

$0 \le K \le M-1$

$E_{s_K} = \overline{S_K^2(t)} = \frac{A^2}{2} \times T_s$ → M-Ary PSK is constant energy scheme

$0 \le K \le M-1$

Binary PSK

$$S(t) = \begin{cases} \overset{s_1(t)}{A\cos 2\pi f_c t} & 0 \le t \le T_b \quad 1T_x \\ \underset{s_2(t)}{-A\cos 2\pi f_c t} & 0 \le t \le T_b \quad 0T_x \end{cases}$$

$E_{s_1} = E_{s_2} \rightarrow$ BPSK is constant energy scheme.

$$S_K(t) = \underset{\downarrow}{\left\{A\sqrt{\frac{T_s}{2}}\cos\phi_K\right\}}\underset{\phi_1(t)}{\left\{\sqrt{\frac{2}{T_s}}\cos 2\pi f_c\right\}} + \underset{\downarrow}{\left\{A\sqrt{\frac{T_s}{2}}\sin\phi_c\right\}}\underset{\phi_2(t)}{\left\{-\sqrt{\frac{2}{T_s}}\sin 2\pi f_c t\right\}}$$

$$\phi_1(t) = \begin{cases} \sqrt{\frac{2}{T_s}}\cos 2\pi f_c t & 0 \le t \le T_s \\ 0 & \text{else} \end{cases}$$

$$\phi_2(t) = \begin{cases} -\sqrt{\frac{2}{T_s}}\sin 2\pi f_c t & 0 \le t \le T_s \\ 0 & \text{else} \end{cases}$$

$\{\phi_1(t), \phi_2(t)\} \rightarrow$ Set of orthonormal basis function.

$$S_K(t) = \left\{A\sqrt{\frac{T_s}{2}}\cos\phi_K\right\}\phi_1(t) + \left\{A\sqrt{\frac{T_s}{2}}\sin\phi_k\right\}\phi_2(t)$$

$0 \le K \le M-1$

M-Ary PSK has constellation diagram of 2 – D.

$$S_0(t) = \left(A\sqrt{\frac{T_s}{2}}\cos\phi_0\right)\phi_1(t) + \left(A\sqrt{\frac{T_s}{2}}\sin\phi_0\right)\phi_2(t)$$

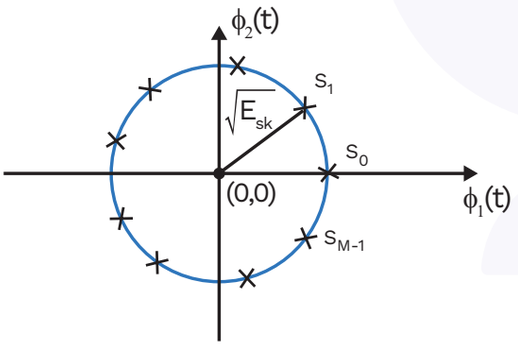
$$d_{S_0} = \sqrt{\frac{A^2T_s}{2}\left(\cos^2\phi_0 + \sin^2\phi_0\right)} = \sqrt{\frac{A^2T_s}{2}} = \sqrt{E_{S_2}}$$

$$S_1(t) = \left(A\sqrt{\frac{T_s}{2}}\cos\phi_1\right)\phi_1(t) + \left(A\sqrt{\frac{T_s}{2}}\sin\phi_1\right)\phi_2(t)$$

$$d_{S_1} = \sqrt{\frac{A^2T_s}{2}\left(\cos^2\phi_1 + \sin^2\phi_1\right)} = \sqrt{\frac{A^2T_s}{2}} = \sqrt{E_{S_1}}$$

Constellation Diagram of M-Ary PSK

Circle with centre (0, 0), radius = $\sqrt{E_{S_K}}$



4-Ary PSK or QPSK: M = 4 N = 2

$$S_K(t) = A\cos\left(2\pi f_c t + \frac{2\pi}{M}K\right) \quad 0 \le t \le T_s$$

$K = 0$ to $M-1$

$$S_K(t) = A\cos\left(2\pi f_c t + \frac{\pi}{2}K\right) \quad 0 \le t \le T_b$$

$K = 0$ to 3

$$S(t) = \begin{cases} A\cos 2\pi f_c t: & 0 \le t \le 2T_b \quad "00" \\ A\cos\left(2\pi f_c t + \frac{\pi}{2}\right): & 0 \le t \le 2T_b \quad "01" \\ A\cos\left(2\pi f_c t + \pi\right): & 0 \le t \le 2T_b \quad "10" \\ A\cos\left(2\pi f_c t + \frac{3\pi}{2}\right): & 0 \le t \le 2T_b \quad "11" \end{cases}$$

Constellation Diagram of QPSK

$$S_K(t) = \left\{A\sqrt{\frac{T_s}{2}}\cos\phi_K\right\}\phi_1(t) + \left\{A\sqrt{\frac{T_s}{2}}\sin\phi_K\right\}\phi_2(t)$$

$T_s = 2T_b$

$K = 0$ to 3

$\phi_0 = 0 \quad \phi_2 = \pi$

$$\phi_K = \frac{2\pi}{M}\times K = \frac{\pi}{2}\times K \longrightarrow \phi_1 = \frac{\pi}{2} \quad \phi_3 = \frac{3\pi}{2}$$

$$S_K(t) = \sqrt{E_S}\cos\phi_K\phi_1(t) + \sqrt{E_S}\sin\phi_K\phi_2(t)$$

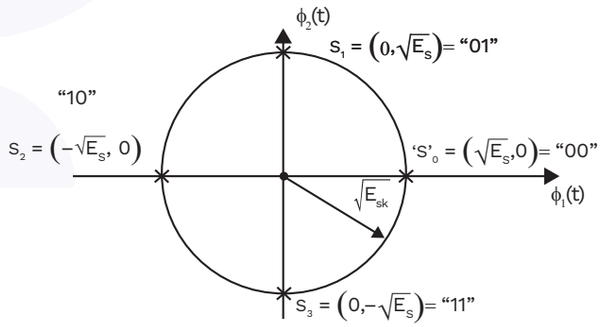
Where $E_s = E_{SK} = \frac{A^2T_s}{2} \longrightarrow$ Energy of any symbol

$S_0 = (t) = \sqrt{E_s}\phi_1(t)$

$S_1 = (t) = \sqrt{E_s}\phi_2(t)$

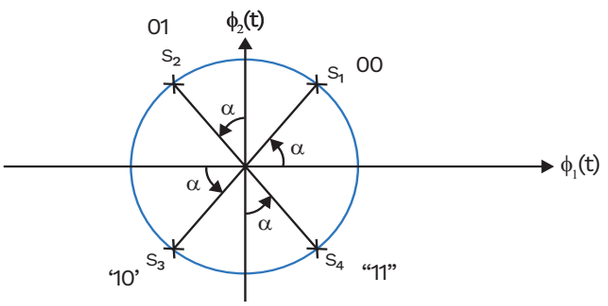
$S_2 = (t) = -\sqrt{E_s}\phi_1(t)$

$S_3 = (t) = -\sqrt{E_s}\phi_2(t)$



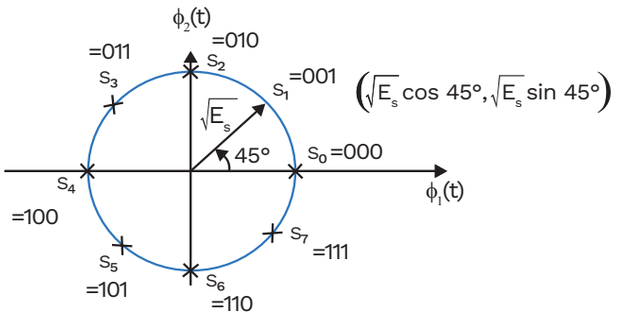
$$S_K(t) = A\cos\left(2\pi f_c t + \frac{2\pi}{M}K + \alpha\right) \quad 0 \le t \le T_s$$

$M = 0$ to $K-1$

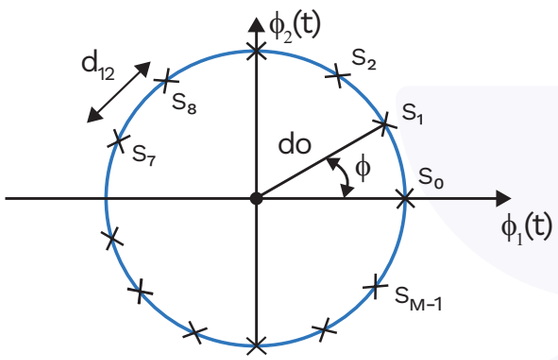


8-Ary PSK

$$\phi_K = \frac{2\pi}{M} \times K = \frac{\pi}{4} \times K$$



Constellation Diagram of M-Ary PSK



Formulae of M-Ary PSK

$M = 2^N$

1. Bit interval $= T_b$
2. Bit rate $= R_b = nf_s$ → Encoder output $\{b_K\}$
3. Symbol duration $T_s = NT_b$
4. Baud or Baud rate or Symbol rate $R_s = \frac{R_b}{N}$
5. Bit energy $E_b = \frac{A^2}{2} \times T_b \quad T_b = \frac{K}{f_c}$
6. Symbol energy

$$E_s = \frac{A^2}{2} \times T_s = \frac{A^2}{2} \times NT_b = NE_b$$

7. $\frac{E_s}{T_s} = \frac{NE_b}{NT_b} = \frac{E_b}{T_b}$
8. Radius of constellation

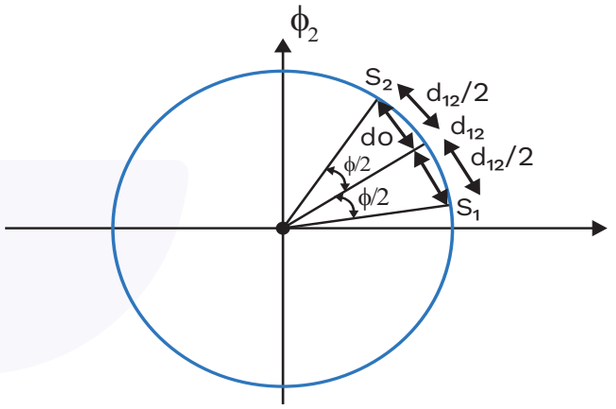
$$d_0 = \sqrt{E_s} = \sqrt{NE_b} = \sqrt{N\frac{A^2}{2}T_b}$$

9. Area of constellation $A = \pi d_0^2 = \pi E_s$
10. Perimeter of constellation
$P = 2\pi d_0 = 2\pi\sqrt{E_s}$
11. Equation of circle: $\phi_1^2(t) + \phi_2^2(t) = d_0^2$
12. Distance between successive symbols on the circle of radius d_0

$$d_{min} = d_{12} = 2d_0 \sin\frac{\phi}{2} \qquad d_0 = \sqrt{E_s}$$

$$\phi = \frac{2\pi}{M}$$

ϕ = Phase difference between consecutive or successive symbols $\phi = \frac{2\pi}{M}$



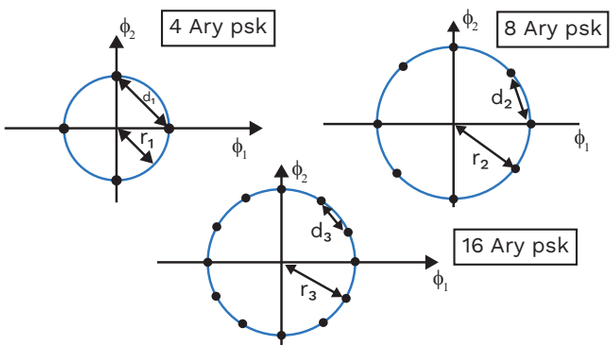
$$\sin\frac{\phi}{2} = \left(\frac{d_{12}/2}{do}\right)$$

Note: "In M-Ary PSK though we can't calculate P_e or P_{se} from constellation as of now but:"

In any constellation:

As $d_{min} \uparrow P_e \downarrow P_{se} \downarrow$

Example:



Relation b/ω r and d =?

Solution:

1. $r_1 = \text{radius} = d_0$

   $d_1 = d_{12}$

   $d_{12} = 2d_0 \sin\frac{\phi}{2}$

   $d_{12} = 2r_1 \sin\frac{\phi}{2} \qquad \phi = \frac{2\pi}{M} = \frac{\pi}{2}$

   $d_{12} = 2r_1 \sin\frac{\pi}{4}$

   $d_{12} = r_1\sqrt{2}$

2. $d_0 = r_2 \qquad d_{12} = 2d_0 \sin\frac{\phi}{2}$

   $d_{12} = d_2$

   $d_2 = 2r_2 \sin\frac{\phi}{2} \qquad \phi = \frac{2\pi}{M} = \frac{2\pi}{8}$

   $d_2 = 2r_2 \sin\frac{\pi}{\theta}$

3. $d_3 = 2r_3 \sin\frac{\pi}{16}$

Calculate Average Symbol Energy

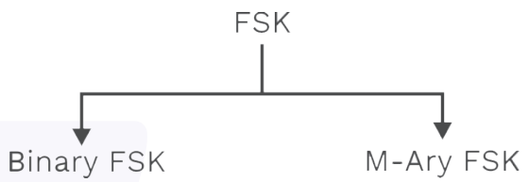
1. $(E_s)_{avg} = \sum_{i=0}^{3} p_i E_{s_i} = \frac{1}{4}\left[E_s + E_s + E_s + E_s\right]$

   $= E_s = NE_b = 2 \times \frac{A^2T_b}{2} = A^2T_b$

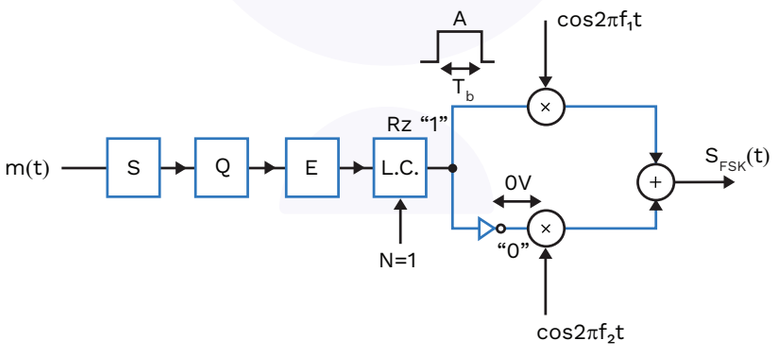
2. $(E_s)_{avg} = \sum_{i=0}^{7} p_i E_{s_i} = \frac{1}{8}\left[E_s + E_s + \ldots\ldots 8\,\text{time}\right]$

   $= E_s = NE_b = 3 \times \frac{A^2T_b}{2} = \frac{3}{2}A^2T_b$

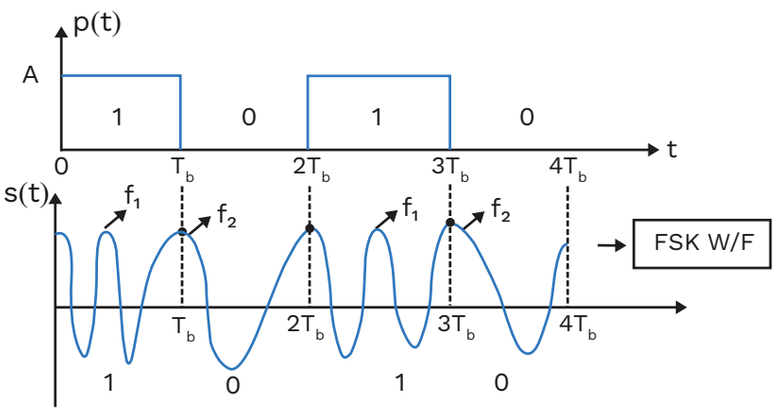
3. $(E_s)_{avg} = E_s = NT_b = 4 \times \frac{A^2T_b}{2} = 2A^2T_b$



Binary FSK:



$$S_{FSK}(t) = \begin{cases} A\cos 2\pi f_1 t & 0 \le t \le T_b : 1T_x \\ A\cos 2\pi f_2 t & 0 \le t \le T_b : 0T_x \end{cases}$$



BASK:

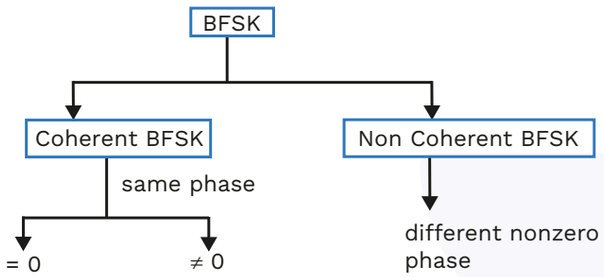
$$S(t) = \begin{cases} S_1(t) = A\cos 2\pi f_c t & 0 \le t \le T_b \quad 1T_x \\ S_2(t) = 0 & 0 \le t \le T_b \quad 0T_x \end{cases}$$

$$\int_0^{T_b} S_1(t)S_2^*(t)dt = 0 \longrightarrow \text{Orthogonal}$$

BPSK:

$$S(t) = \begin{cases} A\cos 2\pi f_c t & 0 \le t \le T_b \quad 1T_x \\ -A\cos 2\pi f_c t & 0 \le t \le T_b \quad 0T_x \end{cases}$$

$$\int_0^{T_b} S_1(t)S_2^*(t)dt \neq 0 \longrightarrow \text{Non-orthogonal}$$



Binary PSK:

Case 1: Coherent FSK with $\phi = 0$.

$$S(t) = \begin{cases} \overset{S_1(t)}{\uparrow} \\ A\cos 2\pi f_c t & 0 \le t \le T_b \quad 1T_x \\ -A\cos 2\pi f_c t & 0 \le t \le T_b \quad 0T_x \\ \underset{S_2(t)}{\downarrow} \end{cases}$$

$E_{s_1} = E_{s_2} \rightarrow$ BPSK is constant energy scheme.

$$= \int_0^{T_b} \frac{A^2}{2}\cos\left(2\pi\left(f_1 + f_2\right)t\right)dt + \int_0^{T_b} \frac{A^2}{2}\cos\left(2\pi\left(f_1 - f_2\right)t\right)dt$$

$$= \frac{A^2 \sin\left(2\pi\left(f_1 + f_2\right)T_b\right)}{4\pi\left(f_1 + f_2\right)} + \frac{A^2 \sin\left(2\pi\left(f_1 - f_2\right)T_b\right)}{4\pi\left(f_1 - f_2\right)}$$

$$\left[2\pi\left(f_1 + f_2\right)T_b = m\pi\right] \cap \left[2\pi\left(f_1 - f_2\right)T_b = n\pi\right]$$

Case 2: Coherent FSK with $\phi \neq 0$.

$$S(t) = \begin{cases} S_1(t) = A\cos(2\pi f_1 t + \phi) : 0 \le t \le T_b : 1Tx \\ S_2(t) = A\cos(2\pi f_2 t + \phi) : 0 \le t \le T_b : 0T_x \end{cases}$$

$$\int_0^{T_b} S_1(t)S_2(t)(dt) = 0$$

$$= \int_0^{T_b} \frac{A^2}{2}\cos[2\pi(f_1 + f_2)t + 2\phi]dt + \int_0^{T_b} \frac{A}{2}\cos[2\pi(f_1 - f_2)t]dt$$

$$= \frac{A^2}{2}\frac{\overbrace{\left[\left\{\sin(2\pi(f_2 + f_2)T_b + 2\phi\right\} - \left\{\sin 2\phi\right\}\right]}^{\sin 2\phi}}{2\pi(f_2 + f_2)} + \frac{A^2 \sin(2\pi(f_1 - f_2)T_b}{4\pi(f_2 - f_2)}$$

$$2\pi(f_2 + f_2)T_b = 2m\pi \ \cap \ 2\pi(f_2 - f_2)T_b = n\pi$$

Condition for Orthogonally

$$\left\{(f_1 + f_2) = mR_b\right\} \cap \left\{(f_1 - f_2) = \frac{nR_b}{2}\right\}$$

Calculation of R_b

$$R_b = HCF(mR_b, nR_b)$$

Case 3: Non-coherent FSK $\rightarrow$ PFSK

$$s(t) = \begin{cases} A\cos\left(2\pi f_1 t + \phi_1\right) & 0 \le t \le T_b \quad 1T_x \\ A\cos\left(2\pi f_2 t + \phi_2\right) & 0 \le t \le T_b \quad 0T_x \end{cases}$$

1. $\left\{\left(f_1 + f_2\right) = mR_b\right\} \cap \left\{\left(f_1 - f_2\right) = nR_b\right\}$
2. $R_b = HCF\left(mR_b, nR_b\right)$

Example: For a bit rate of 8 Kbps, the best possible values of the transmitted frequencies in a coherent Binary FSK system are:

(A) 16 K and 20 K

(B) 20 K and 32 K

(C) 20 K and 40 K

(D) 32 K and 40 K

$R_b = 8Kbps$

$FSK \rightarrow$ Orthogonal FSK

Solution:

Binary PSK:

$$S(t)=\begin{cases} \overset{S_1(t)}{\uparrow} \\ A\cos 2\pi f_c t & 0\le t\le T_b \quad 1T_b \\ -A\cos 2\pi f_c t & 0\le t\le T_b \quad 0T_x \\ \underset{S_2(t)}{\downarrow} \end{cases}$$

$E_{S_1}=E_{S_2}\rightarrow$ BPSK is constant energy scheme

Example: In a digital communication system employing FSK, the 0 and 1 bit are represented by sine waves of 10 kHz and 25 kHz, respectively. These waveforms will be orthogonal for a bit interval of:

$T_b=?$

(A) 45μ sec

(B) 200μ sec

(C) 50μ sec

(D) 250μ sec

Case 1:

Coherent $\quad \phi=0$

$f_1=10K \qquad f_2=25K$

$$f_1+f_2=\frac{mR_b}{2}\rightarrow mR_b=70K$$

$$f_1-f_2=\frac{nR_b}{2}\rightarrow nR_b=30K$$

$$R_b=HCF(70K,30K)=10K$$

$$T_b=\frac{1}{10K}=100\mu\text{ sec}$$

Case 2:

Coherent $\quad \phi\neq 0$

$$f_1+f_2=mR_b \qquad f_1-f_2=\frac{nR_b}{2}$$

$$mR_b=35K \qquad nR_b=30K$$

$$R_b=HCF(35K,30K)=5K$$

$$T_b=\frac{1}{R_b}=200\mu\text{ sec}$$

Case 3:

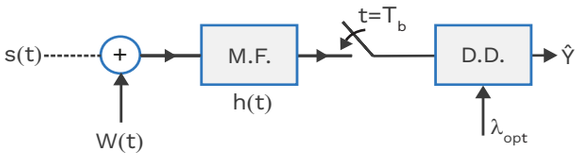
$$f_1+f_2=mR_b \qquad f_1-f_2=nR_b$$

$$mR_b=35K \qquad nR_b=15K$$

$$R_b=HCI\left(35K,15K\right)=5K$$

$$T_b=\frac{1}{R_b}=200\mu\text{ sec}$$

Matched Filter Based R_x



1. $h(t)=S_1\left(T_b-t\right)-S_2\left(T_b-t\right) \qquad 0\le t\le T_b$

2. $P_e=Q\left[\sqrt{\frac{E_d}{2N_0}}\right]$

$$E_d=\int_0^{T_b}\left(S_1(t)-S_2(t)\right)^2dt=E_{S_1}+E_{S_2}-2\underset{\downarrow 0}{\int_0^{T_b}}S_1(t)\,S_2(t)dt$$

Orthogonal FSK

$$\longrightarrow E_d=E_{S_1}+E_{S_2}$$

$$S(t)=\begin{cases} A\cos 2\pi f_1 t & 0\le t\le T_b \longrightarrow \quad T_b=\frac{K}{f_1}=\frac{m}{f_2} \\ A\cos 2\pi f_2 t & 0\le t\le T_b \longrightarrow \quad E_{S_1}=E_{S_2}=\frac{A^2}{2}T_b \end{cases}$$

$$E_d=\frac{A^2T_b}{2}+\frac{A^2T_b}{2}=A^2T_b$$

$$P_e=Q\left[\sqrt{\frac{A^2T_b}{2N_0}}\right]$$

Correlator Receiver

$$S(t)=\begin{cases} A\cos 2\pi f_1+0\le t\le T_b:1T_x \\ A\cos 2\pi f_2+\le t\le T_b:0T_x \end{cases} \qquad \text{M=2}$$

$$N\le 2\rightarrow N=2\quad \phi_1(t)=\begin{cases} \sqrt{\frac{2}{T_b}}\cos 2\pi f_1 t & 0\le t\le T_b \\ 0 & \text{else} \end{cases}$$

$$\phi_2(t) = \begin{cases} \sqrt{\frac{2}{T_b}} \cos 2\pi f_2 t & 0 \le t \le T_b \\ 0 & \text{else} \end{cases}$$

$$s = \{\phi_2(t), \phi_2(t)\} \rightarrow O.N.B.F$$

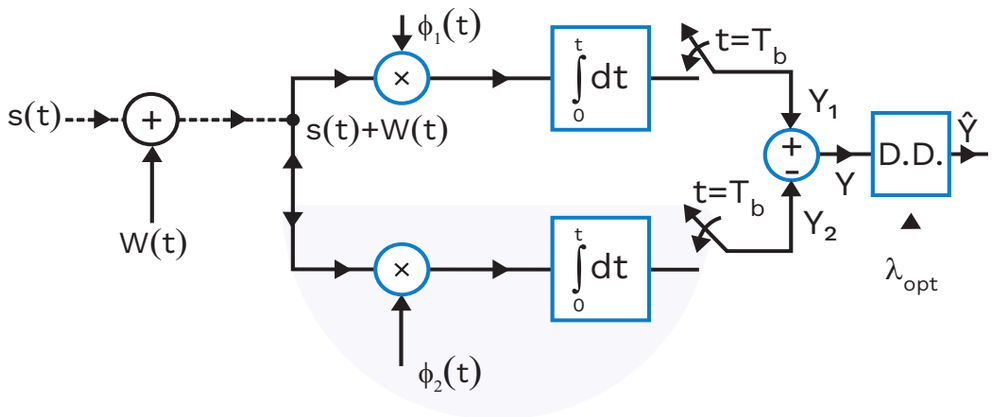
$$s(t) = \begin{cases} A\sqrt{\frac{T_b}{2}}\phi_1(t) & : 0 \le t \le T_b \quad 1T_x \\ A\sqrt{\frac{T_b}{2}}\phi(t) & : 0 \le t \le T_b \quad 0T_x \end{cases}$$

$$Y_2 = \int_0^{T_b} A\sqrt{\frac{T_b}{2}}\phi_1(t)\phi_2(t)dt + \int_0^{T_b} W(t)\phi_2(t)dt$$

$$Y = Y_1 - Y_2 = A\sqrt{\frac{T_b}{2}} + N_1 - N_2 \quad 1Tx$$

When 0 Tx:

$$S_2(t) = a\sqrt{\frac{T_b}{2}}\phi_2(t) : 0 \le t \le T_b$$



$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi f_1 t$$

$$\phi_2(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi f_2 t$$

When 1 is T_{xed}

$$s_1(t) = A\sqrt{\frac{T_b}{2}}\phi_1(t) : 0 \le t \le T_b$$

$$S_1(t) + W(t) = A\sqrt{\frac{T_b}{2}}\phi_1(t) + W(t) \quad 0 \le t \le T_b$$

Upper Segment

$$Y_1 = \int_0^{T_b} A\sqrt{\frac{T_b}{2}}\phi_1^2(t)dt + \int_0^{T_b} \phi_2(t)W(t)dt$$

$$Y_1 = A\sqrt{\frac{T_b}{2}} + N_1$$

Lower Segment

$$S_2(t) + W(t) = A\sqrt{\frac{T_b}{2}}\phi_2(t) + W(t) : \ 0 \le t \le T_b$$

Upper Segment

$$Y_1 = \int_0^{T_b} A\sqrt{\frac{T_b}{2}}\phi_2(t)\phi_1(t)dt + \int_0^{T_b} \phi_1(t)W(t)dt$$

$$Y_1 = N_1$$

Lower Segment

$$Y_2 = A\sqrt{\frac{T_b}{2}} + N_2$$

$$Y = Y_1 - Y_2 = -A\sqrt{\frac{T_b}{2}} + N_1 - N_2 \ 0Tx$$

Input of DD

$$Y = \begin{cases} A\sqrt{\frac{T_b}{2}} + N_1 - N_2 & : 1T_x \\ -A\sqrt{\frac{T_b}{2}} + N_1 - N_2 & : 0T_x \end{cases}$$

$$N_1 \begin{cases} \nearrow & E(N_1) = 0 \\ \searrow & \sigma^2_{N_1} = \dfrac{N_0}{2} \end{cases}$$

Same for N_2

$$E[Y] = \begin{cases} A\sqrt{\dfrac{T_b}{2}} & 1T_x \\ -A\sqrt{\dfrac{T_b}{2}} & 0T_x \end{cases}$$

$$\sigma^2_y = \begin{cases} \sigma^2_{N_1} + \sigma^2_y - 2\operatorname{cov}(N_1 N_2)\, 1T_x \\ \sigma^2_{N_1} + \sigma^2_{N_2} - 2\operatorname{cov}\left(\underset{\to 0}{N_2}\, N_2\right) 0T_x \end{cases}$$

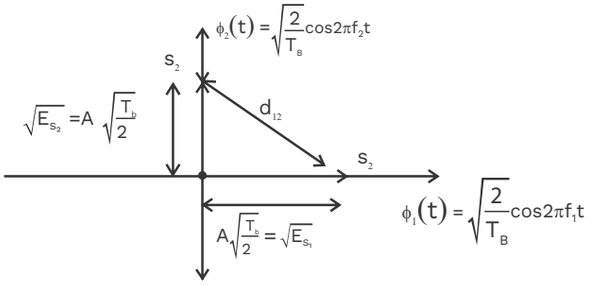
Let N_1, N_2 independent

$$P_e = Q\left[\frac{\mu_1 - \mu_2}{2\sigma_y}\right]$$

$$P_e = Q\left[\sqrt{\frac{A^2 T_b}{2N_0}}\right]$$

Geometrical Representation of BFSK

$$S(t) = \begin{cases} A\sqrt{\dfrac{T_b}{2}}\phi_1(t): & 0 \le t \le T_b \quad : 1T_x \\ +A\sqrt{\dfrac{T_b}{2}}\phi_2(t); & 0 \le t \le T_b \quad : 0T_y \end{cases}$$



$$P_e = Q\left[\sqrt{\frac{d_{12}^{\,2}}{2N_0}}\right] = Q\left[\sqrt{\frac{4^2 T_b}{2N_0}}\right]$$

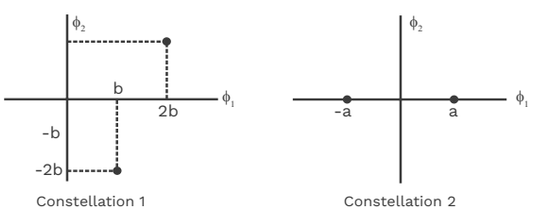
Binary PSK:

$$s(+)\begin{cases} \overset{\nearrow s_1(+)}{A\cos 2\pi f_c +} & 0 \le t \le T_b \quad 1T_x \\ \underset{s_2 \swarrow (+)}{-A\cos 2\pi f_c +} & 0 \le t \le T_b \quad 0T_x \end{cases}$$

$Es_1 = Es_2 \rightarrow$ BPSK is constant energy scheme

Test Your Understanding

1. Consider the signal constellations 1 and 2, as shown below proposed for an equiprobable binary communication system over the AWGN channel.



Which of the following is true?

(A) Constellation 1 is advantageous in the sense that it is two-dimensional while constellation 2 is one-dimensional.

(B) For a = 2 and b = 1, constellation 2 contributes larger probability of error.

(C) For 2a = $b\sqrt{10}$, both constellations contribute equal probability of error.

(D) For a = 1 and = 2, constellation 1 contribute larger probability of error.

Solution: (C)

2. A message signal has strictly no frequency component than 20khz, is sampled at Nyquist rate, quantized into Q levels, and then the resulting digital data is transmitted using M-ray PSK with raised cosine pulses of roll the of factor α. A Channel with a 120 kHz Bandwidth is available for transmission. For Q = 16 and M = 4, the largest permissible value α is____.

Solution: 0.5

M-Ary FSK: N bits are grouped together so that M = 2^Nsymbols or sinusoids of duration $T_s = NT_b$ are generated having:

→ Same amplitude.

→ Same phase.

→ Different Freq.

$$S_k(t) = A\cos\left[2\pi f_k t\right] : o \le t \le T_s : k = 0\,to\,M-1$$

$$\overline{S_k^2}(t) = \frac{A^2T_s}{2}\ 0 \le k \le M-1 \rightarrow \text{costant energy scheme.}$$

$$4-Ary\ FSK : \boxed{T_s = 2T_b} \qquad M = 4 \qquad N = 2$$

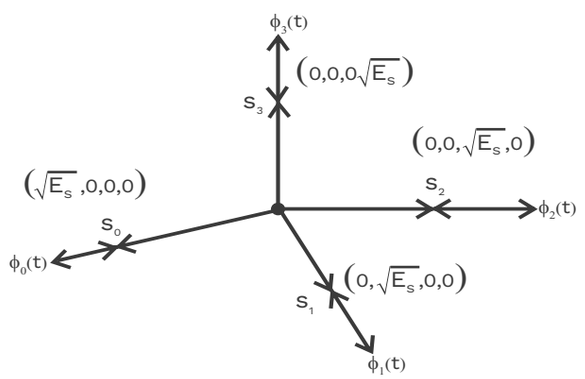
$$S(t) = \begin{cases} A\cos 2\pi f_0 t : & 0 \le t \le 2T_b \\ A\cos 2\pi f_1 t : & 0 \le t \le 2T_b \\ A\cos 2\pi f_2 t : & 0 \le t \le 2T_b \\ A\cos 2\pi f_3 t : & 0 \le t \le 2T_b \end{cases}$$

$$\phi_i(t) = \begin{cases} \sqrt{2/_{Ts}}\cos 2\pi f_i & 0 \le t \le 2t_b \quad i = 0,1,2,3 \\ 0 & else\ where \end{cases}$$

$$S(t) = A\begin{cases} A\sqrt{\frac{Ts}{2}}\phi_0(t) : & 0 \le t \le 2T_b \quad 00 \\ A\sqrt{\frac{Ts}{2}}\phi_1(t) : & 0 \le t \le 2T_b \quad 01 \\ A\sqrt{\frac{Ts}{2}}\phi_2(t) : & 0 \le t \le 2T_b \quad 10 \\ A\sqrt{\frac{Ts}{3}}\phi_0(t) : & 0 \le t \le 2T_b \quad 11 \end{cases}$$

$$A\sqrt{\frac{T_s}{2}} = \sqrt{E_s}$$

Constellation Diagram



Some important notes:

1. Constellation diagram of M-Ary ASK can be 1-D or 2-D.
2. Constellation diagram of 2-Ary PSK can be 1-D or 2-D.
3. Constellation diagram of M-Ary PSK will always be 2-D (Circle).
4. Constellation diagram of 2-Ary FSK is 2-D.
5. Constellation diagram of M-Ary FSK is M-D.
6. Relation between Q function and Compli-mentary error function.

$$\boxed{Q(x) = \frac{1}{2}erfc\left(\frac{X}{\sqrt{2}}\right)} \quad or \quad \boxed{erfc(x) = 2Q\left(\sqrt{2X}\right)}$$

7. Average bit energy:

$$\boxed{BASK:}\ (E_b)avg = p_0Es_0 + p_1Es_1$$

$$\frac{1}{2}\times 0 + \frac{1}{2}\times\left\{\frac{A^2T_b}{2}\right\}$$

$$\boxed{(E_b)avg = \frac{A^2T_b}{4} = E_b}$$

$$\boxed{BPSK:}\quad (E_b)avg = p_0Es_0 + p_1Es_1$$

$$= \frac{1}{2}\times\frac{A^2T_b}{2} + \frac{1}{2}\times\frac{A^2T_b}{2}$$

$$\boxed{(E_b)avg = \frac{A^2T_b}{2}}$$

8. P_e for Binary Bandpass scheme with 1 channel:

SCHEME	P_e	P_e
BASK	$Q\left[\sqrt{\frac{A^2T_b}{4N_0}}\right]$	$Q\left[\sqrt{\frac{(E_b)avg}{N_0}}\right]$
BPSK	$Q\left[\sqrt{\frac{A^2T_b}{N_0}}\right]$	$Q\left[\sqrt{\frac{2(E_b)avg}{N_0}}\right]$
BFSK(orthogonal)	$Q\left[\sqrt{\frac{A^2T_b}{2N_0}}\right]$	$Q\left[\sqrt{\frac{(E_b)avg}{N_0}}\right]$

SCHEME	P_e	P_e
BASK	$Q\left[\sqrt{\frac{A^2T_b}{4N_0}}\right]$	$\frac{1}{2}erf_c\left[\sqrt{\frac{A^2T_b}{8N_0}}\right]$
BPSK	$Q\left[\sqrt{\frac{A^2T_b}{N_0}}\right]$	$\frac{1}{2}erf_c\left[\sqrt{\frac{A^2T_b}{2N_0}}\right]$
BFSK(orthogonal)	$Q\left[\sqrt{\frac{A^2T_b}{2N_0}}\right]$	$\frac{1}{2}erf_c\left[\sqrt{\frac{A^2T_b}{4N_0}}\right]$

9. P_e for K Independent, identically distributed AWGN channel:

Scheme	P_e
BASK	$Q\left[\sqrt{\frac{KA^2T_b}{4N_0}}\right]$
BPSK	$Q\left[\sqrt{\frac{KA^2T_b}{N_0}}\right]$
BFSK(orthogonal)	$Q\left[\sqrt{\frac{KA^2T_b}{2N_0}}\right]$

10. Properties of constellation:
    a) Distance from origin to point
    $S_1 = \sqrt{E_{s_1} = d_0}$
    b) Min. distance between 2 points = $\sqrt{Ed = d_{12}}$
    c) For 2-point constellation:
    $$P_e = Q\left[\sqrt{\frac{d_{12}^2}{2N_0}}\right] = \frac{1}{2}erfc\left[\sqrt{\frac{d_{12}^2}{4N_0}}\right]$$
    d) For multipoint constellation.

$P_e = ?$
As $d_{12} \uparrow \quad P_e \downarrow$
$d_{12} = distance\ between\ consecutive\ point$

"Amplitude: Phase shift keying"
(M-point QAM)

$$S_i(t) := r_i \cos(2\pi f_c t + \theta_i) : 0 \le t \le T_s$$
$$i = 0 \ \ to\, M-1$$

CASE 1 $r_i = constant \rightarrow M - Ary\ PSK$
$Q \rightarrow variable$

CASE 2 $r_i = variable \rightarrow M - Ary\ ASK$
$Q \rightarrow constant$

M-Ary FSK: N bits are grouped together so that $M = 2^N$ symbols or sinusoids of duration $T_s = NT_b$ are generated having:

- Same amplitude
- Same phase.
- Different Freq.

Non coherent Rx:

(1) BASM → ?
(2) BPSK → ? → DPSK (Differental PSK)
(3) BFK → ?

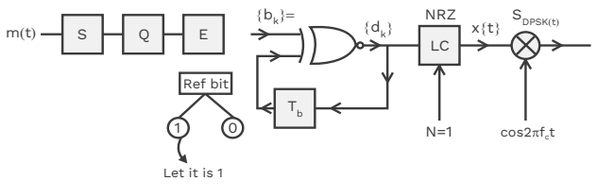
Differential PSK (DPSK):

COHERENTR$_x$

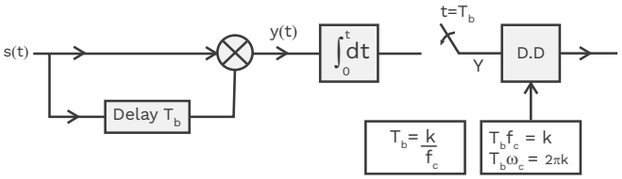
BPSK :→ Correlator Based R_x or
MATCHEDFILITERBASED R_x

$$P_e = Q\left[\sqrt{\frac{A2T_b}{N_0}}\right] \xrightarrow[ERROR]{PHASE} P_e = Q\left[\sqrt{\frac{A2T_b}{N_0}\cos^2\theta}\right]$$

DPSK T_x:



DPSK R_x:



$$y(t) = S(t)S(t - T_b)$$

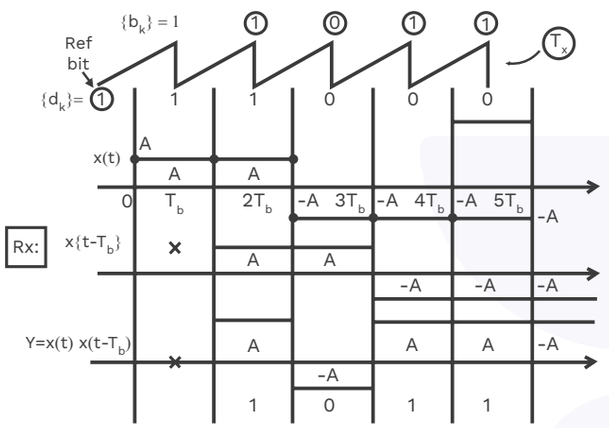
$$y(t) = x(t)\,x(t-T_b)\cos\omega_c t \cos\left(\omega_c t - \underbrace{\omega_c T_b}_{\uparrow\ 2\pi k}\right)$$

$$y(t) = x(t) + x(t-T_b)\cos^2\omega_c t$$

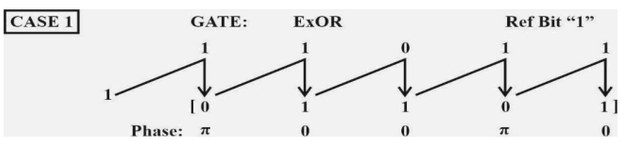
$$Y = \int_0^{T_b}\frac{1}{2}x(t)\,x(t-T_b)\,dt + \int_0^{T_b}\frac{1}{2}\underbrace{\cos 2\omega_c t}_{\to 0}\ \text{dt}$$

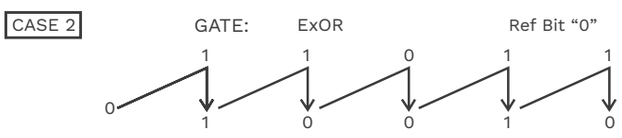
$$Y = \frac{T_b}{2}x(t)\ \ x(t-T_b)$$

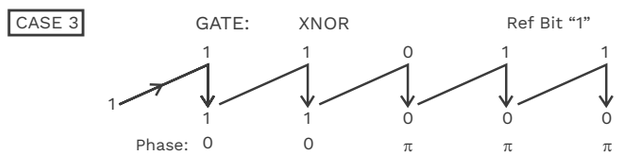
$$Y\alpha\ \ x(t)x(t-T_b)$$

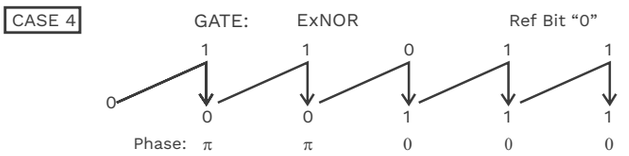


3. The message bit sequence of the DPSK modulator is 11011. The carier phase during the reception of the first two bits are π, π. The carier phase of remaining 3 bits are:



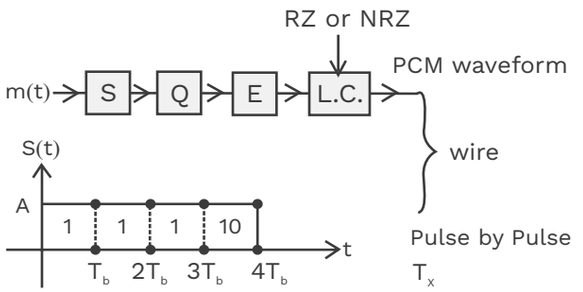




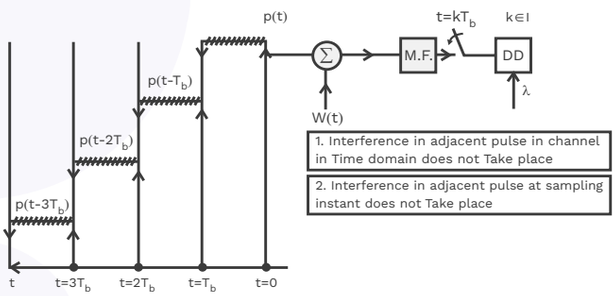


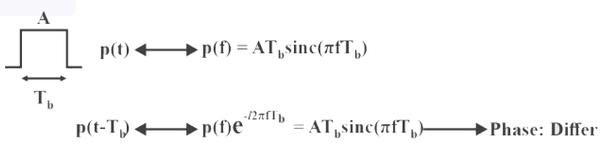
Solution: 0° 0° 0°

Intersymbol Interference (ISI)

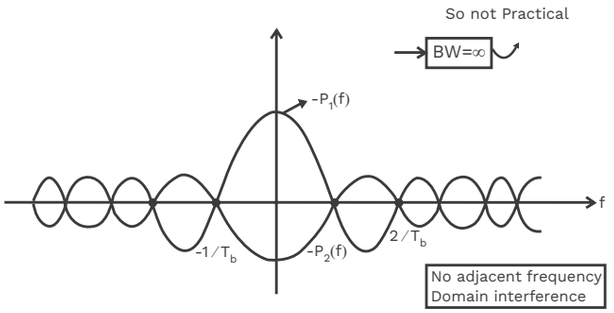


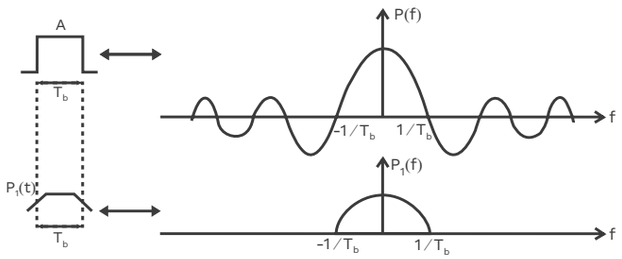
Channel in time domain:



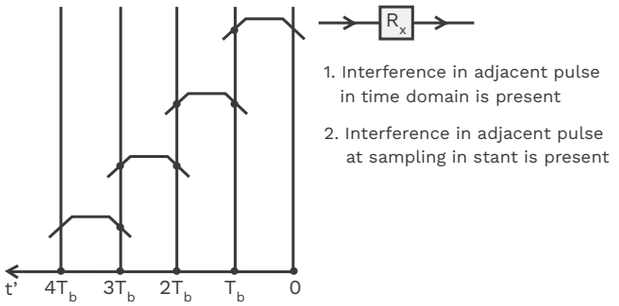


Channel in frequency domain:





Channel domain:



1. Interference in adjacent pulse in time domain is present
2. Interference in adjacent pulse at sampling in stant is present

1. If RZ-NRZ pulse shaping is used to represent bit 1 or bit 0. They require infinite bandwidth.
2. If RZ-NRZ pulses are Band limited, they expand in time domain, reseating in interference of adjacent pulses both in the channel and at sampling in an instant known as ISI.
3. ISI is a time domain phenomenon.
4. ISI leads to improper detection of pulses, thereby increasing P_e
5. ISI comes into the picture because RZ-NRZ coding is used to represent Bit '1' Bit '0'.
6. To remove ISI, pulse shaping is used.

Removal of ISI at sampling instance:

1. Every pulse should have finite BW.
2. Every pulse should not interfere in the time domain, at least at the sampling instant.

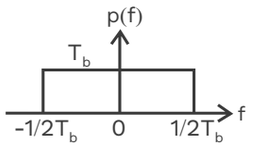
Pulse shaping:

1. Sinc pulse shaping.
2. Raised cosine pulse shaping.

Sinc pulse shaping:

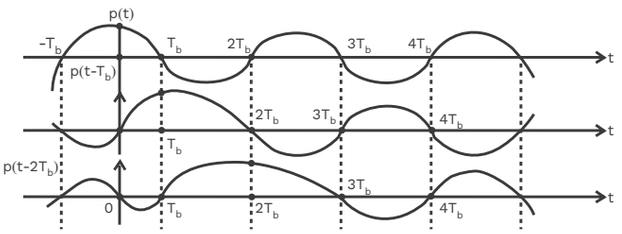
$$'1' = p(t) = \text{sinc}\left(\frac{t}{T_b}\right) \rightarrow$$ Zero crossovers multiple of T_b

$$p(t) = \text{sinc}\left(\frac{t}{T_b}\right) \longleftrightarrow$$

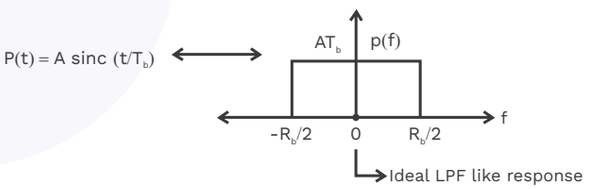


Bandlimited

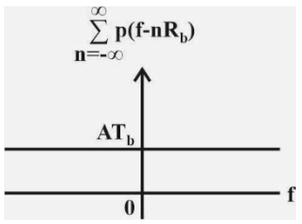
Channel in Time domain



1. ISI can never be removed from channel.
2. ISI can be removed only at the sampling instant by using proper pulse shaping.
3. One such pulse shape is Sinc with cross-overs with T_b.



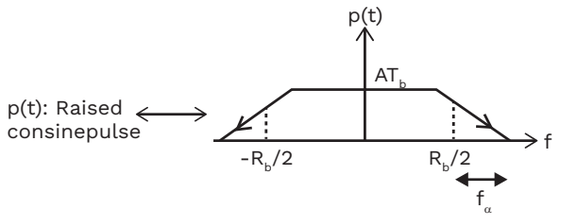
4. Calculate $\sum_{n=-\infty}^{\infty} p(f - nR_b)$



Solution:

$$p(t) = A\,\text{sinc}\left(\frac{t}{T_b}\right) \leftrightarrow p(f)$$

$$\sum_{n=\infty}^{\infty} P(f - nR_b) = AT_b = \text{constant}$$



Channel BW for zero ISI when "Raise cosine pulse shaping" is used.

$$(Bw) \geq \frac{R_b}{2} + f_\alpha$$

$f_\alpha \rightarrow ExessB.W.$

$\frac{R_b}{2} \rightarrow Ideas\ BW$

$$f_\alpha = \alpha \frac{R_b}{2}$$

α : Roll off factor :

$$\# \boxed{Roll\ off\ factor = \alpha = \frac{ExcessB.W.}{Ideal\ BW} = \frac{f_\alpha}{(R_b)_{/2}}}$$

$$\# \boxed{\%ExcessBW = \%\alpha = \frac{f_\alpha}{(R_b)_{/2}} \times 100\%}$$

$$\boxed{o \leq \alpha \leq 1}$$

Points To Remember

1. To remove ISI at sampling instant for 2-Ary scheme.

$p(t) \leftrightarrow p(f)$

$\Sigma p(f - nR_b) = AT_b = \text{constant}$

$R_b = nf_s$: Bit rate

2. To remove ISI at sampling instant for M-Ary Scheme:

$$\sum_{n=-\infty}^{\infty} P(f - nRs) = \text{constant}$$

R_s : Symbol Rate

Formula of Bandwidth

Case 1 : $BaseBand + 2Ary + NRz$

$(BW) \geq R_b$

Case 2 : $BaseBand + MAry + NRz$

$(BW) \geq R_s$

$T_s = NT_b$

$R_s = \frac{R_b}{\log_2 M}$

Case 3 : $Bandpass + 2Ary + NRz$

$(BW) \geq 2R_b \rightarrow {}^{BASK}_{BPSK} \rightarrow$ Not for BFSK

Case 4 : $Bandpass + MAry + NRz$

$(BW) \geq 2R_s \nearrow M\ Ary\ ASK \searrow M\ Ary\ PSK \rightarrow$ Not for M Ary FSK

Case 5 : $BaseBand + \text{Sin}c + 2Ary$

$(BW) \geq \frac{R_b}{2} \rightarrow$ Zero ISI

Case 6 : $BaseBand + \text{Sin}c + MAry$

$(BW) \geq \frac{R_s}{2} \rightarrow$ Zero ISI

Case 7 : $Bandpass + \text{Sin}c + 2Ary$

$(BW) \geq R_b \rightarrow$ Zero ISI

Case 8 : $Basepass + \text{Sin}c + MAry$

$(BW) \geq R_s \rightarrow$ Zero ISI

Case 9 : $BaseBand + Raised\ \text{cosine} + 2Ary$

$(BW) \geq \frac{R_b}{2}(1+\alpha) \rightarrow$ Zero ISI

Case 10 : $BaseBand + Raised\ \text{cosine} + MAry$

$BW \geq \frac{R_s}{2}(1+\alpha) \rightarrow$ Zero ISI

Case 11 : $Bandpass + Raised\ \text{cosine} + 2Ary$

$BW \geq R_s(1+\alpha) \rightarrow$ Zero ISI

Case 12 : $Bandpass + Raised\ \text{cosine} + MAry$

$BW \geq R_s(1+\alpha) \rightarrow$ Zero ISI

Note:

BaseBand $\left(BW \geq \frac{R_b}{2}\right) \rightarrow$ Default

$\left\{\begin{matrix} PCM \\ DPCM \end{matrix}\right. \xrightarrow{BW}$ Pulse shape: Sinc

Bandpass $\xrightarrow{BW}$ Pulse shape: Rz/NRz

$\left\{\begin{matrix} PCM \\ DPCM \end{matrix}\right.$ $BASK: (BW \geq 2R_b)$

5. Determine the bit rate that can be T_{xed} through a 4 kHz voice band telephone channel (Bandpass) if the following modulation schemes are used.

1. BPSK.
2. 4-Ary PSK.
3. 8 Point QAM.

Solution:

$R_b = nf_s$

BW of Bandpass $= 2 \times BW$ of Baseband

$f_{max} = 2KHz$

$f_s = 2f_{max} = \frac{4Ksampler}{sec}$

Bandwidth of baseband:

1. $BPSK: R_b = nf_s \rightarrow R_s = nf_s$
2. $QPSK: R_b = nf_s \rightarrow R_s = \frac{nf_s}{2}$
3. 8 point QAM: $R_b = nf \rightarrow R_s = \frac{nf_s}{3}$

$Encoder \rightarrow \frac{nbits}{sample}$

(3) Determine maximum bit rate:

$Bandpass + (Rz - NRz) + M - Ary$

$BPSK: (BW) \geq 2R_b \rightarrow R_b \leq \frac{(BW)}{2}$

$(R_b)_{max} = 2kbps.$

$QPSK: (BW) \geq 2R_s \rightarrow (BW) \geq \frac{2R_b}{\log_2 M}$

$(R_b)_{max} = 4kbps.$

8 point QAM: $(BW) \geq 2R_s \rightarrow$

$(BW) \geq \frac{2R_b}{\log_2 M} \rightarrow (R_b)_{max} = 6kbps$

(4) Calculate maximum bit rate for zero ISI.

$BPSK: (BW) \geq R_S \quad (R_b)_{max} = 4kbps$

$QPSK: (BW) \geq R_S \quad (R_b)_{max} = 8kbps$

8 point QAM: $(BW) \geq R_S \quad (R_b)_{max} = 12kbps$

6. A speech signal is sampled at a rate of 8 kHz and encoded into PCM format using 8bits/sample. The PCM data is transmitted through an AWGN baseband channel via M-level PAM (Pulse-amplitude modulation). Determine BW required for transmission for:

1. M=4
2. M=8
3. M=16

Solution:

Baseband + Zero ISI + M-Ary

$(BW) \geq \frac{R_s}{2}(1+\alpha) \quad \alpha = 0 \quad R_b = nf_s = 64kbps$

$(BW)_{min} = \frac{R_b}{2\log_2 M} = \frac{32kbps}{\log_2 M}$

1. $M = 4 \quad (BW)_{min} = 16KHz$
2. $M = 8 \quad (BW)_{min} = \frac{32}{3}KHz$
3. $M = 16 \quad (BW)_{min} = 8KHz$

Note:

$(BW) \geq \frac{R_s}{2}(1+\alpha)$

Bandpass+MAry $\left|\begin{matrix} \xrightarrow{No\ ISI} & BW \geq R_s(1+\alpha) \\ \xrightarrow{RZ+NRZ} & BW \geq 2R_s \end{matrix}\right.$

Exercise

1. An M-level PSK modulation scheme is used to transmit independent binary digits band over a bandpass channel with a bandwidth 100 kHz. The bit rate is 200 kbps, and the system characteristic is a raised cosine spectrum with 100% excess bandwidth. The minimum value of M is _____.

Solution: 16

2. A speech signal is sampled at 8 kHz and encoded into PCM format using 8 bits per sample. The PCM data is transmitted through a baseband channel via 4-level PAM. The minimum bandwidth (in kHz) required for transmission is _____.

Solution: 16

3. A digital communication system transmits a block of N bits. The probability of error in decoding a bit is α. The error event of each bit is independent of the error events of the other bits. The received block is declared erroneous of at least one of its bits is decoded wrongly. The probability that the received block is erroneous is:

(A) $1-(1-\alpha)^N$

(B) α^N

(C) $N(1-\alpha)$

(D) $1-\alpha^N$

Solution: A

4. The bit rate of a digital communication system is R kbits/s. The modulation used is 32 QAM. The minimum bandwidth required for ISI-free transmission is:

(A) R/10 Hz

(B) R/10 kHz

(C) R/5 Hz

(D) R/5 kHz

Solution: D

5. A source emits bit 0 with probability $\frac{1}{3}$ and bit 1 with probability . The emitted bits are communicated to the receiver. The receiver decides on either 0 or 1 based on the received value R. It is given that the conditional density functions of R as:

$$f_{R/0}(r)=\begin{cases}\frac{1}{4}, & -3\le r\le 1\\ 0, & \textit{Otherwise}\end{cases}$$

$$f_{R/1}(r)=\begin{cases}\frac{1}{6}, & -1\le r\le 5\\ 0, & \textit{Otherwise}\end{cases}$$

The minimum decision error probability is:

(A) 0

(B) $\frac{1}{12}$

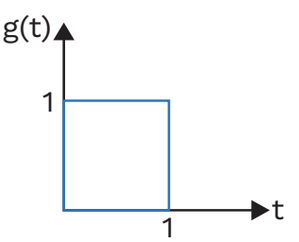
(C) $\frac{1}{9}$

(D) $\frac{1}{6}$

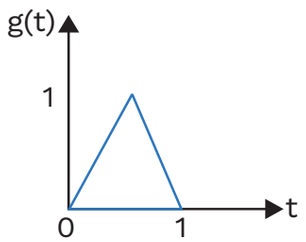
Solution: D

6. Consider a binary, digital communication system which uses pulses g (t) and –g (t) for transmitting bits over an AWGN channel. If the receiver uses a matched filter, which one of the following pulses will give the minimum probability of bit error?

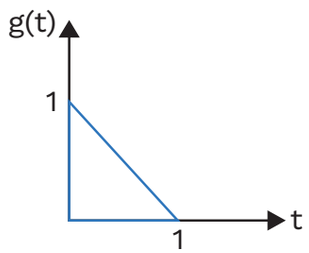
(A)



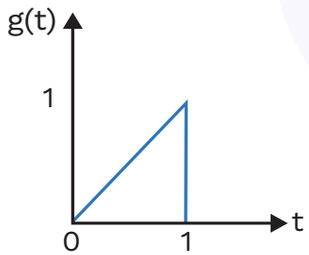
(B)



(C)



(D)



Solution: (A)

7. A random variable 'X' takes values –0.5 and 0.5 with probabilities $\frac{1}{4}$ *and* $\frac{3}{4}$, respectively. The noisy observation of X is $Y = X + Z$, where Z has uniform probability density over the interval (–1, 1). X and Y are independent. If the MAP rule-based detector outputs $\hat{X}$ as:

$$\hat{X} = \begin{cases} -0.5 & Y < \alpha \\ 0.5 & Y \geq \alpha \end{cases}$$

Then the value of α (accurate to two decimal places) is _____.

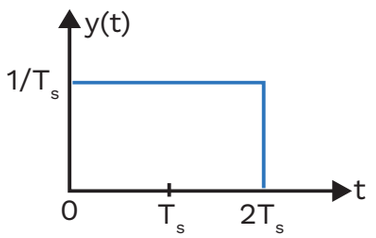
Solution: (–0.5)

8. A binary baseband digital communication system employs the signal.

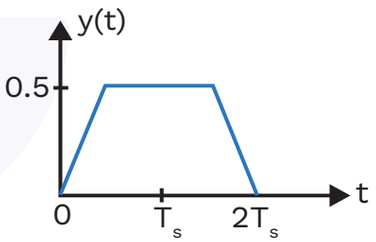
$$p(t)\begin{cases} \frac{1}{\sqrt{T_s}} & 0 \leq t \leq T_s \\ 0, & otherwise \end{cases}$$

For transmission of bits, the graphical representation of the matched filter output y (t) for this signal will be:

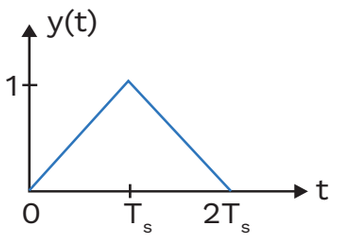
(A)



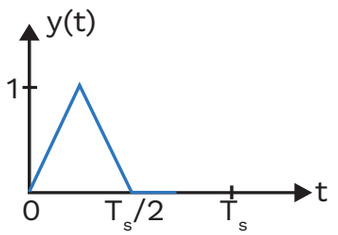
(B)



(C)



(D)



Solution: (C)

Solutions

1. Given: BW = 100 kHz, $f_b = 200\,kbps$

$\alpha = 1 or\ 100\%$

The bandwidth of M-ary PSK is given by,

$$BW = \frac{2f_b}{N}$$

Relationship between number of bits and number of symbols is given by,

$$M = 2^N$$

$\log_2 M = N$

With raised cosine spectrum, bit rate f_b is replaced by $\frac{(1+\alpha)R_b}{2}$ and bandwidth is given by,

$$BW = \frac{2}{N}\frac{R_b}{2}(1+\alpha)$$

$$BW = \frac{R_b}{log_2 M}(1+\alpha)$$

$$100\times10^3 = \frac{200\times10^3}{\log_2 M}(1+1)\left[\because R_b = f_b\right]$$

$$\log_2 M = \frac{400\times10^3}{100\times10^3} = 4$$

$$M = 2^4 = 16$$

Hence, the minimum value of M is 16.

2. Given: $f_s = 8\,kHz$

n=8 bits/sample, M = 4

Minimum bandwidth is given by,

$$(BW)_{min} = \frac{R_b}{2log_2 M} = \frac{R_b}{4} = \frac{nf_s}{4}$$

$$(BW)_{\min} = \frac{8\times8}{4} = 16\,kHz$$

Hence, the minimum bandwidth required for transmission is 16 kHz.

3. Given probability of error in decoding a single bit = α, N bit error = α^N

Then probability of no error for single bit will be = $1-\alpha$

Block size = N bits



The received block is declared erroneous, if at least one of bits is decoded wrongly.

Probability of received correct block

$$=(1-\alpha)(1-\alpha).....N-times=(1-\alpha)^N$$

Then, the probability of received block is erroneous = $1-(1-\alpha)^N$

Hence, the correct option is (a).

4. Given: M = 32, $R_b = R\ kbits/s$

For M-ary QAM, transmission bandwidth is given by,

$$BW = \frac{2R_b}{\log_2 M}\ where, R_b = Bit\ rate$$

For ISI free transmission, bandwidth is given by,

$$BW = \frac{2R_b}{2}(1+\alpha)\times\frac{1}{\log_2 M}$$

For minimum bandwidth, α = 0

$$BW = \frac{R_b}{\log_2 M} = \frac{R}{\log_2 32} = \frac{R}{5}kHz$$

Hence, the correct option is (d).

5. Solution:

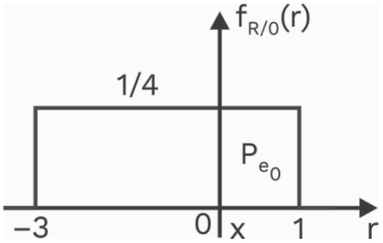
Given: $P(0)=\frac{1}{3}, P(1)=\frac{2}{3}$

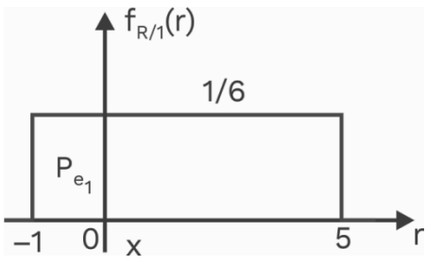
The conditional density function of R as

$$f_{R/0}(r) = \begin{cases}\frac{1}{4}, & -3\le r\le 1\\ 0, & otherwise\end{cases}$$

$$f_{R/1}(r) = \begin{cases}\frac{1}{6}, & -1\le r\le 5\\ 0, & otherwise\end{cases}$$

PDFs for the received bits are as follows





Let us assume that the decision boundary is at r = x.

Let P_{e_0} = Probability of error if '0' was transmitted

P_{e_1} = Probability of error if '1' was transmitted.

$$P_{e_0} = \int_x^1 f_{R/0}(r)\,dr = \int_x^1 \frac{1}{4}\,dr$$

$$P_{e_0} = (1-x)\frac{1}{4}$$

and $$P_{e_1} = \int_{-1}^{x} f_{R/1}(r)\,dr = \int_{-1}^{x} \frac{1}{6}\,dr$$

$$P_{e_1} = (x+1)\frac{1}{6}$$

Probability of error is given by,

$$P_e = P_{e_0}P(0) + P_{e_1}P(1)$$

$$P_e = (1-x)\frac{1}{4}\times\frac{1}{3} + (x+1)\frac{1}{6}\times\frac{2}{3}$$

$$P_e = \frac{(1-x)}{12} + \frac{(x+1)}{9}$$

$$P_e = \frac{7+x}{36}$$

P_e is minimum if x is minimum but $-1 \le x \le 1$.

So, P_e is minimum if x = –1.

$$P_e = \frac{1}{6}$$

Hence, the correct option is (d).

6. Probability of error of matched filter is

$$P_e = Q\left[\frac{1}{2}\sqrt{\left(\frac{S}{N}\right)_0}\right]$$

Q(x) decreases with increase in x and vice-versa.

Thus for having minimum error we have to select signal with maximum energy.

Energy of signal 1,

$$E_1 = \int_0^1 1^2 dt = 1$$

Energy of signal 2,

$$E_2 = \int_0^{0.5} (2t)^2\,dt + \int_{0.5}^{1} (-2t+2)^2\,dt$$

$$E_2 = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

Energy of signal 3,

$$E_3 = \int_0^1 (-t+1)^2\,dt = \frac{1}{3}$$

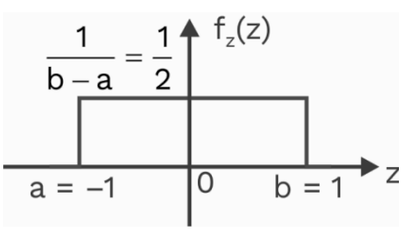
Energy of signal 4,

$$E_4 = \int_0^1 (t)^2\,dt = \frac{1}{3}$$

Signal 1 has maximum energy hence it will provide minimum probability of error.

Hence, the correct option is (a).

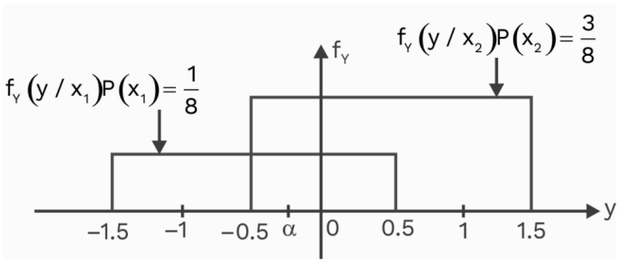
7. Given: Z is uniformally distributed random variable in the range –1 < z < 1



Probabilities of random variable X being –0.5 and + 0.5 as,

$$P(x_1) = P(x = -0.5) = \frac{1}{4}$$

$$P(x_2) = P(x = 0.5) = \frac{3}{4}$$



Probability of error is given by,

$$P_e = P\left(\frac{0.5}{-0.5}\right)P(-0.5) + P\left(\frac{0.5}{0.5}\right)P(0.5)$$

$$P_e = \frac{1}{2}(0.5 - \alpha) \times \frac{1}{4} + \frac{1}{2} \times \frac{3}{4} \times (\alpha + 0.5)$$

$$P_e = \frac{1}{4} \times \frac{1}{2} \times (0.5 - \alpha) + \frac{1}{2} \times \frac{3}{4}(\alpha + 0.5)$$

$$P_e = \frac{1}{16} - \frac{\alpha}{8} + \frac{32}{28} + \frac{3}{16} = \frac{1+\alpha}{4}$$

Threshold voltage lies in the range

$-0.5 \le \alpha \le 0.5$

At $\alpha = -0.5, P_e = 0.125$

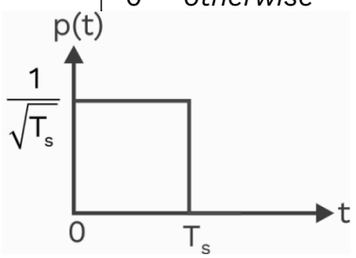
At $\alpha = 0, P_e = 0.25$

At $\alpha = 0.5, P_e = 0.375$

Therefore, P_e will be minimum when $\alpha = -0.5$

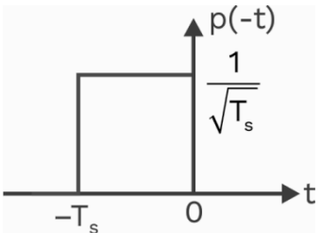
Hence, the value of α is – 0.5.

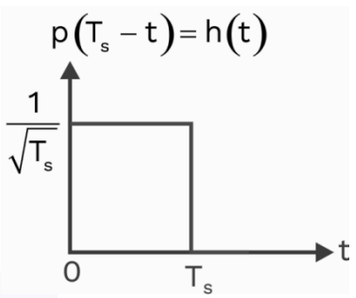
8. Given: $p(t) = \begin{cases} \frac{1}{\sqrt{T_s}} & 0 \le t \le T_s \\ 0 & otherwise \end{cases}$



Impulse response of matched filter

$$h(t) = p(T_s - 1)$$





After folding the signal if we advance the signal it actually gets delayed.

Output of matched filter

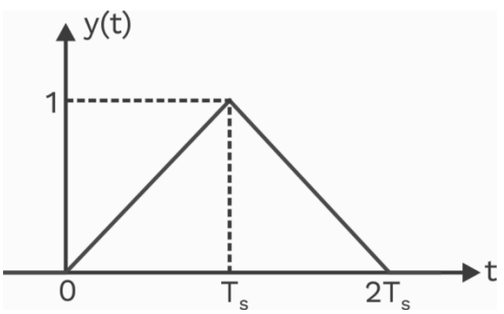
$$y(t) = p(t) \otimes h(t)$$

Note: Convolution of two rectangular signals of same duration results in triangular signal.

Amplitude of output signal

= Energy of the input signal.

$$= \left(\frac{1}{\sqrt{T_s}}\right)^2 \times T_s = 1$$



Hence, the correct option is (c).